REVIEW

A scoping and bibliometric review of deep learning techniques in breast cancer imaging: mapping the landscape and future directions

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Received: 4 April 2024 / Accepted: 26 March 2025 / Published online: 13 May 2025
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Abstract
In recent years, there has been a rise in interest in the application of deep learning (DL) to breast cancer identification. This work intends to deliver a review demonstrating new DL applications for identifying and categorizing breast cancer and offers a bibliometric analysis of advancements in this field. A comprehensive search between January 1, 2016, and February 20, 2023, was done on the Scopus database to perform the bibliometric analysis of DL applications in breast cancer detection. The statistics on publications, journals, authors, nations, and keywords were created using VOSviewer software. Moreover, Medline (through PubMed), Web of Science (WoS), and Scopus were used for the scoping review (from January 1, 2016, to December 24, 2022); all retrieved titles were checked for eligibility requirements by the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) methodology. Since 2016, the frequency of papers being recovered has grown, and in 2022, a significant number of citations were retrieved. This paper gives a general overview of the various DL methodologies and tailored architectures for the detection, categorization, and segmentation of breast cancer. Performance of the approaches is assessed using accuracy, sensitivity, specificity, area under curve (AUC), F1 score, Dice similarity coefficient (DSC), and intersection over union (IoU); results with high accuracy are frequently obtained at the expense of sensitivity.

Keywords Breast cancer · Deep learning · Medical image · Bibliometric review

Abbreviations

CNN	Convolutional neural network
DCNN	Deep convolutional neural network
DCE-MRI	Dynamic contrast-enhanced magnetic resonance imaging
OCT images	Optical coherence tomography
CEDM	Contrast-enhanced digital mammography
DBT images	Digital breast tomosynthesis
BI-RADS	Breast imaging reporting and data system
GBM	Gradient boosting algorithm
RBCNN	Region-based convolutional neural network
MIAS database	Mammographic image analysis society
ReLU	Rectified linear unit
RPN	Region proposal network

https://doi.org/10.1007/s00521-025-11215-4

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DDSM	Digital database for screening mammography
BUS	Breast ultrasound
GLCM	Gray-level co-occurrence matrix
ADUNet	Attention dense UNet
CMMD	The Chinese mammography database
BCSDBT	Breast cancer screening–digital breast tomosynthesis
DWI	Diffusion-weighted imaging
CADx	Computer-aided diagnosis
ALN	Axillary lymph node
YOLO	You-only-look-once
FrCN	Full resolution convolutional network
CBIS-DDSM	Curated breast imaging subset of DDSM
ABUS	Automated breast ultrasound
DV-DCNN	Dual-view deep convolutional neural network
mtNEt	Multitask net
SN-Net	Stitchable neural networks
S-DPN	Stacked deep polynomial network
SC-FC-LSTM-FCN	Spatial channel long short-term memory fully convolutional neural networks
FFS-CNN	Feature fusion Siamese neural network
NAC	Neoadjuvant chemotherapy
LSTM	Long short-term memory
AM-MSP-cGAN	Attention mechanism and multi-scale pooling conditional generative adversarial network
BCDR	Breast cancer digital repository
CSwin-PNet	A CNN-Swin Transformer combined pyramid network
BTEC-Net	Breast tumor ensemble classification network
RFS-UNet	Residual feature selection UNet
CBAM	Convolutional block attention module
LDA	Linear discriminant analysis
FFDM	Full-field digital mammography dataset
CEDM	Contrast-enhanced digital mammography
SD-CNN	Shallow–deep convolutional neural network
GBT	Gradient boosting tree

1 Introduction

Breast cancer is the most common cancer diagnosed in women [1, 2]. As one of the common causes of death [3], this disease ranks second among the causes of death caused by cancer in women [4]. In the USA, 12% of women will receive a breast cancer diagnosis at some point in their lives. In 2017, the USA discovered more than 250,000 new cases of breast cancer [5]. In the UK, around 55,000 women are diagnosed with breast cancer each year, which equates to one every 10 min. One woman out of eight may be diagnosed with breast cancer during her lifetime [6]. Early detection of breast cancer can significantly increase the chance of survival [7], provide more flexible treatment options for patients, improve quality of life [8], and reduce costs [9, 10].

Medical images, which make up a big part of studies, are seen as one of the most important ways to figure out a person's problem. The generated images must undergo significant processing before they can be used as raw data without analysis and interpretation since the information in these images is hidden in the pixels [11]. The most

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efficient way to diagnose breast cancer is by looking at medical imaging [12]. Digital mammography (DM), ultrasound (US), magnetic resonance imaging (MRI), microscopic images (histology), and infrared thermography are a few examples of the medical imaging techniques utilized for diagnosis [13]. Medical imaging methods, especially DM, are well-known and suitable ways to screen, diagnose, and measure breast density in a timely way [14]. However, mammography-based diagnosis by radiologists is likely to be false positive, leading to unnecessary imaging and tumor biopsy [15].

Due to the heterogeneity of breast density and the low contrast of mammography, feature selection and manual feature removal are computationally challenging and time-consuming [16]. Unfortunately, it is impossible to completely eliminate the possibility of errors when interpreting medical images; among the elements that compromise the precision of diagnosis are the variety of tissues and cells, a scarcity of medical professionals with specific specialties, lengthy workdays, eye fatigue, and a lack of focus [17]. In order to mitigate these issues, a lot of research has been done recently to automate the interpretation of medical images. Computer-aided image analysis can be effective in the early diagnosis of diseases [17] and significantly improve the reproducibility and accuracy of diagnosis [9, 18].

Clinical disease diagnosis, treatment evaluation, and the detection of defects in various body organs, including the eye, lungs, brain, breast, and stomach, all depend heavily on medical imaging. Medical imaging describes a variety of methods used to examine the human body to identify, monitor, or treat diseases. For the purposes of disease diagnosis and monitoring, each technology type delivers specific information about the area of the body being examined or handled, the illness, injury, or success of the medical therapy [19]. Classifying the location, size, and characteristics of the organ under study is the goal of medical imaging investigations, which has been recognized as an efficient way to sort through the vast amount of data to find what is relevant.

To identify the majority of disorders, various researchers concentrated heavily on the creation and interpretation of medical images [7, 20]. The identification of a pathological lesion, the therapeutic care of the patient, and the inference of several medical conditions are all made possible thanks to medical imaging. The medical business, including the processing of medical images, has been significantly impacted by recent developments in machine learning (ML) and artificial intelligence (AI) [21, 22].

AI has gained popularity in recent years with the idea of employing computers to carry out either simple or sophisticated jobs typically performed by humans [23]. By utilizing the ML process, these AI computers are sometimes able to conduct tasks that are impossible for humans to complete or perform considerably more effectively [24, 25]. Due to the rapid development of DL and ML algorithms, AI has demonstrated exceptional success in the analysis of medical images. These accomplishments have improved the capacity to address challenging real-world issues and evaluate visual data [26, 27].

Artificial neural networks (ANNs), which are widely used in the field of rapid tumor diagnosis by analyzing medical images and using image processing and ML techniques such as segmentation, data compression, image enhancement, and noise removal [28, 29], can improve the accuracy of breast cancer diagnosis. For extracting nonlinear, entangled, and representative features from high-dimensional, large-scale data, DL uses artificial neural networks [30].

Multilayer networks are used in DL, a branch of ML that draws inspiration from the human brain's capacity for learning, to assess complicated patterns in raw image input data [21, 31]. Although the architecture of these deep networks is somewhat complex, they are computationally more powerful than other machine learning techniques like support vector machines, logistic regression, random forests, linear regression, and k-nearest neighbor [31].

DL as a powerful tool has had many successes in image analysis and relatively challenging tasks such as X-ray image analysis, computed tomography (CT) scans, and MRIs, as well as ML such as tissue classification in histopathology and histology images [31–33].

Techniques based on DL have become popular in the health sector, including breast cancer detection. AI and DL have been suggested as imaging techniques more efficient than other imaging techniques in various studies [2]. In addition, DL is useful in the field of breast imaging, and diagnostic support systems based on DL are becoming more common [34].

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A deep neural network (DNN) can also reduce the amount of human error and the pressure on the physicians in predicting breast lesions [24] and improve the automatic detection of the mass, location of the mass, feature learning, and classification [8, 10]. To clarify the motivation behind and goal of this study, several review articles in this area are provided below. These explain the shortcomings of the prior work.

1.1 Motivation and aim

Several review studies with different objectives have been conducted on the use of DL or ML in the field of breast cancer.

A systematic evaluation of computer-aided diagnosis (CAD) of breast cancer utilizing several imaging modalities and ML approaches was presented by Yasin et al. in 2018. The performance measures that are frequently employed to compare breast cancer image diagnoses were also tested in this work, along with a variety of classifiers and feature extraction techniques. The fact that this research was conducted in an earlier era and was only limited to academic and scientific publications was one of its limitations [13].

A critical review article on DL applications in breast tumor identification utilizing ultrasound and mammography images was published in 2020 by Jiménez-Gaona et al. The drawback of this study was that some algorithms and their outputs were unavailable due to concerns about proprietary intellectual property. The main results of the classification process showed that the new DL-CAD methods provided a useful way to automatically pullout features using a self-learning network, which improved the accuracy of the classification. Better structures, larger datasets that address class imbalance issues, and improved optimization techniques, however, are constantly needed [35].

The DL architectures utilized for various imaging approaches; model performances for various tasks like classification and segmentation, and research goals were all thoroughly examined by Madani et al. in 2022. The findings of this study demonstrated that DL techniques, despite their exceptional performance, still have several issues that must be resolved before they may finally have an impact on clinical procedures. Before utilizing DL to its full potential in breast cancer clinical imaging practice, ethical concerns regarding the explainability and interpretability of these systems should be taken into account, in addition to the difficulties [36].

Nasser et al. (2023) conducted a comprehensive study of DL-based methods for diagnosing breast cancer using genomic and histopathological imaging data. According to the systematic review's findings, the most widely used technique for performance evaluation is accuracy metrics, and the convolutional neural network (CNN) is the most widely used and accurate model for diagnosing breast cancer. The study's findings were presented at the annual meeting of the American Association of University Women. Only articles in the English language were included in the search. When evaluating study articles, primary databases were considered, but it is likely that other digital libraries with relevant studies were missed [37].

Mridha et al. (2021) conducted a systematic and analytical review that investigated DL in breast cancer diagnosis, focusing on six aspects of breast cancer diagnosis model architecture, datasets, image preprocessing, breast cancer imaging modality, performance metrics, and research directions [38].

To assess the diagnostic effectiveness of DL algorithms for the early diagnosis of breast and cervical cancer, Peng Xue et al. (2022) performed a meta-analysis. The findings demonstrated that DL algorithms may perform as well as human doctors when used to diagnose breast and cervical cancer using medical imaging. Evidence-based, industry-standard protocols for study and reporting procedures are required to raise the caliber of DL research [39].

Murtaza et al. (2020) systematically and critically focused on the classification of breast cancer using multimodal medical imaging methods through advanced artificial DNN approaches. This study showed that mammography and histopathological images are mostly used to classify breast cancer. In addition, about 55% of the selected studies used public datasets, and the rest used proprietary datasets. Finally, this review presents ten open research challenges for future researchers [40].

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The unique uses of ML and DL technologies for breast cancer diagnosis and classification, as well as breast cancer classification employing multimodality medical imaging, were thoroughly reviewed by Houssein et al. (2021). The categories and obstacles for diagnosing breast cancer in the future were finally discussed [30].

Considering the serious challenges mentioned regarding the diagnosis of breast tumors as well as the problems of performing the process manually, the examination of breast tumors by intelligent algorithms can help physicians make a definitive diagnosis using this method. Therefore, researchers in this study intend to conduct a scoping and bibliometric review to fully investigate DL algorithms for breast cancer. The purpose of this research is to investigate the applications of algorithms as well as the obstacles and problems of developing these DL approaches. In addition to these, determining the research gaps and future suggestions in each study is one of the main goals of the present systematic review. The main points covered in the relevant literature are included in Table 1.

This review seeks to locate different studies in the literature that classify breast cancer using a variety of medical imaging modalities based on deep and ML methods. The main contributions of this study, then, are trying to address the following queries:

(i) How is the bibliometric evaluation of current retrieval publications (Scopus retrievals)?
(ii) What imaging techniques have been used to categorize, screen, and segment breast cancer?
(iii) What datasets are used in the medical image classification models?
(iv) Which DL approaches are currently used to categorize or segment breast cancer based on medical imaging modalities?
(v) Which CNN architecture is employed to categorize breast cancer?
(vi) What measures are employed to assess the effectiveness of the classification models?
(vii) What is the main application of the technique in each study?
(viii) What are the limitations and obstacles to implementation?

Table 1 Comparison of discussed review articles about ML applications for breast cancer diagnosing based on images

References	Type of review	The most impressive features discussed in the reviews						
		Description of image modalities	Description of deep learning approaches (i.e., feature selection methods, architecture, training, and testing dataset)	Evaluation metrics of each technique	Type of applications of machine learning techniques	Applied databases in papers	Limitations of each reviewed paper	Future directions for each reviewed paper
[30]	Comprehensive review	*	*	*	*	*	*	*
[41]	Comparative review	–	*	*	*	*	*	–
[35]	Critical review	*	*	*	*	*	–	–
[13]	Systematic review	*	*	*	*	*	–	–
[40]	Systematic and critical review	*	–	*	*	*	*	–
[36]	Systematic review	*	–	–	–	*	–	–
[37]	Systematic review	*	–	*	*	*	–	–
[38]	Systematic review	*	*	–	–	*	*	*
[39]	Systematic review and meta–analysis	–	*	*	–	*	–	–

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(ix) What are the research gaps or future recommendation paradigms in each study?

In Sect. 2, the methodology is explored. Section 3 presents the results and related analysis. In Sect. 4, discussion is presented. Finally, in Sect. 5, the conclusion is addressed.

2 Methods

According to the Arksey and O'Malley framework [42], there are six stages of conducting a scoping review that this research authors follow: The research topic is stated in detail, and then relevant literature is found, studies are chosen, the data are mapped out, the findings are summarized, synthesized, and reported, and professional guidance is sought.

Also, a qualitative analysis method was employed to compile the examined papers and produce fresh, noteworthy findings. The Preferred Reporting Items for Systematic Review and Meta-analysis (PRISMA) statement [43] provides the foundation for this scoping review's reporting.

Bibliometrics quantitatively examines the results of published studies on a subject to identify recurrent themes or patterns, taking into consideration the sample, the country, the methodology, etc. The VOS viewer software was applied in this work to construct a graphical mapping of the bibliometric data using several indicators, including the co-occurrence of keywords, bibliographic coupling, the co-citation of journals, etc.

2.1 Design of search strategy

Since Scopus contains the majority of the literature on DL techniques for breast cancer diagnosis based on image processing, the authors of this study used its data. Scopus was employed to select the meta-data information for the sample of research because it is a well-known citation database and is frequently used as a reference database for many bibliometric analyses. Additional export options provided by Scopus include references, journals, language, keywords, abstracts, complete citation details, author affiliations, and publication year. Likewise, 2000 publications can be exported at once. Furthermore, Scopus appears to be a trustworthy, authoritative database with broad coverage of the literature. The key descriptive elements of the chosen papers were compiled using a standardized form made in Microsoft Excel. In order to define the search string, the research questions were connected to a list of keywords that were taken from studies on DL applications for breast cancer diagnosis. Besides this, synonyms and alternative spellings were improved iteratively along with the keywords. The keyword set was improved after being tested in various databases.

On February 20, 2023, a bibliographic database based on academic publications on the application of DL techniques for breast cancer diagnosis or segmenting based on image processing was created. The following keywords were applied in a thorough bibliographic search in the Scopus database: ("breast cancer"), ("deep learning"), and ("Img*").

Next, the research team searched three databases to conduct a scoping review. To find papers from January 1, 2016, to December 24, 2022, electronic search tactics were used with Medline (via PubMed), Web of Science (WoS), and Scopus. This scoping review's search method included keywords with Medical Subject Headings (Mesh) terminology relating to "Breast Tumor," "deep learning," and "image processing." In Table 2, a complete list of the keywords and terms used in the PubMed database search strategy is provided.

A manual search was also conducted using two techniques: (1) looking through the references of the linked papers, and (2) utilizing the Google Scholar search engine. This was done to prevent prejudice and to make sure that all pertinent articles were looked for.

We performed the bibliometric analysis in the second month of 2023, that is why we searched the database again on this date.

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Table 2 Search strategy for PubMed

PubMed	(((“Breast Tumor”[Title/Abstract] OR “Mammary Cancer”[Title/Abstract] OR “Malignant Neoplasm of Breast”[Title/Abstract] OR “Breast Malignant Neoplasm”[Title/Abstract] OR “Malignant Tumor of Breast”[Title/Abstract] OR “Breast Malignant Tumor”[Title/Abstract] OR “Cancer of *Breast”[Title/Abstract] OR “Cancer of the Breast”[Title/Abstract] OR “Human Mammary Carcinomas”[Title/Abstract] OR “Human Mammary Carcinoma”[Title/Abstract] OR “Human Mammary Neoplasm”[Title/Abstract] OR “Breast Carcinoma”[Title/Abstract] OR “Breast Carcinomas”[Title/Abstract]) OR (“Breast Neoplasms”[MeSH Terms])) AND ((((“deep Learning”[Title/Abstract] OR “Unsupervised Machine Learning”[Title/Abstract] OR “Supervised Machine Learning”[Title/Abstract] OR “Neural Networks, Computer”[Title/Abstract] OR “Computational Neural Network”[Title/Abstract] OR “Convolutional neural network”[Title/Abstract] OR CNN[Title/Abstract]) OR (Deep learning[MeSH Terms])) OR (Neural Networks, Computer[MeSH Terms]))) AND ((Image Processing, Computer-Assisted[MeSH Terms]) OR (imag*[Title/Abstract])) From* 2016 to 2022

2.2 Eligibility criteria for the scoping review

Following are our established inclusion and exclusion criteria that were used to screen the chosen academic articles:

2.2.1 Inclusion requirements

Due to the proper research question being defined in the determination criteria, more relevant information may be found in the literature. Therefore, the PCC technique (Population or participants, Concept, Context) was used to construct the research questions and inclusion criteria for a qualitative review study.

- *Population*: MRI, US, and X-ray-oriented images of breast tissue.
- *Concept*: Improving breast cancer diagnosis or segmenting using DL
- *Context*: Applying different DL algorithms to medical images.

2.2.2 Exclusion criteria

(1) Studies that have not used MRI, US, or X-ray-oriented images of breast tissue,
(8) Non-English papers,
(3) Articles that are reviews, book chapters, letters, reports, and technical reports.
(4) Manuscripts that are in the preprint phase were also excluded.

2.3 Data collection process and data items for the scoping review

After removing duplicate entries, 1886 papers were found in our search of scientific databases. After filtering the citations, reviewers established several criteria for inclusion and exclusion. Five independent reviewers (SR, EE, NR, GA, and MZ) determined the primary categories of chosen publications and synthesized the essential elements of chosen citations throughout the data extraction step. SR, LSH, MN, and HA approved the papers' essential specifications. Under the direction of LSH, SR, and MN, five reviewers meticulously assessed the titles and abstracts of the citations to locate relevant papers independently. Based on the study questions and particular objectives, all titles and abstracts were evaluated to pick relevant publications. A random sample of papers was reviewed by another reviewer (SR). A total of 230 publications was chosen to proceed to the full-text review stage because they matched our inclusion criteria. Four reviewers carefully read the complete texts of the pertinent papers (SR, GA, NR, and EE). Finally, 135 citations were found to be pertinent, and each paper's key details were recorded in an Excel spreadsheet. Based on the predetermined classification, four writers (SR, NR, GA, EE, and MZ) independently extracted and examined the study characteristics for each manuscript. Two

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authors went back over the data to determine the best extraction and come to a consensus. There was no disagreement between the reviewers. Figure 1 lists the main categories and citations that have been reviewed.

3 Results

3.1 Bibliometric analysis and visualization

After searching the Scopus database for bibliographic analysis (from 2016 to early 2023), 1855 documents were retrieved and then analyzed. This work demonstrates and enriches the body of bibliometrics literature. One of the most popular and precise ways to analyze and study large volumes of scientific data is bibliometric analysis. It is an effective methodology for analyzing huge datasets. Bibliometrics allows for several levels of analysis: macro (in our case, nearly 1900 pieces of evidence provided over only seven years: distribution of papers by subject area, distribution of papers by year, and distribution of results by the document type); moderate (in this study, examining the contributions of authors and countries); and micro (counting the occurrences of each word).

3.1.1 Distribution of document types in terms of years

The original document type was found to be the most frequently used among the eleven various types of papers that were found. From a total of 1855 documents retrieved from 2016 to early 2023, 1443 were original article documents (77.7%), 219 were conference papers (11.8%), 124 were reviews (6.6%), 24 were conference reviews (1.2%), 20 were editorials (1%), 10 were book chapters (0.54%), 10 were notes (0.53%), two were short surveys (0.10%), one was data paper (0.05%), one was letter (0.05%), and one was retracted type (0.05%). As can be seen in Fig. 2, the frequency of retrieved documents has increased since 2016, and in 2022, the number of retrieved results has been extremely significant. This increasing trend can be attributed to the increasing interest of scientists in the field of applying artificial intelligence, especially DL, in the field of diagnosis.

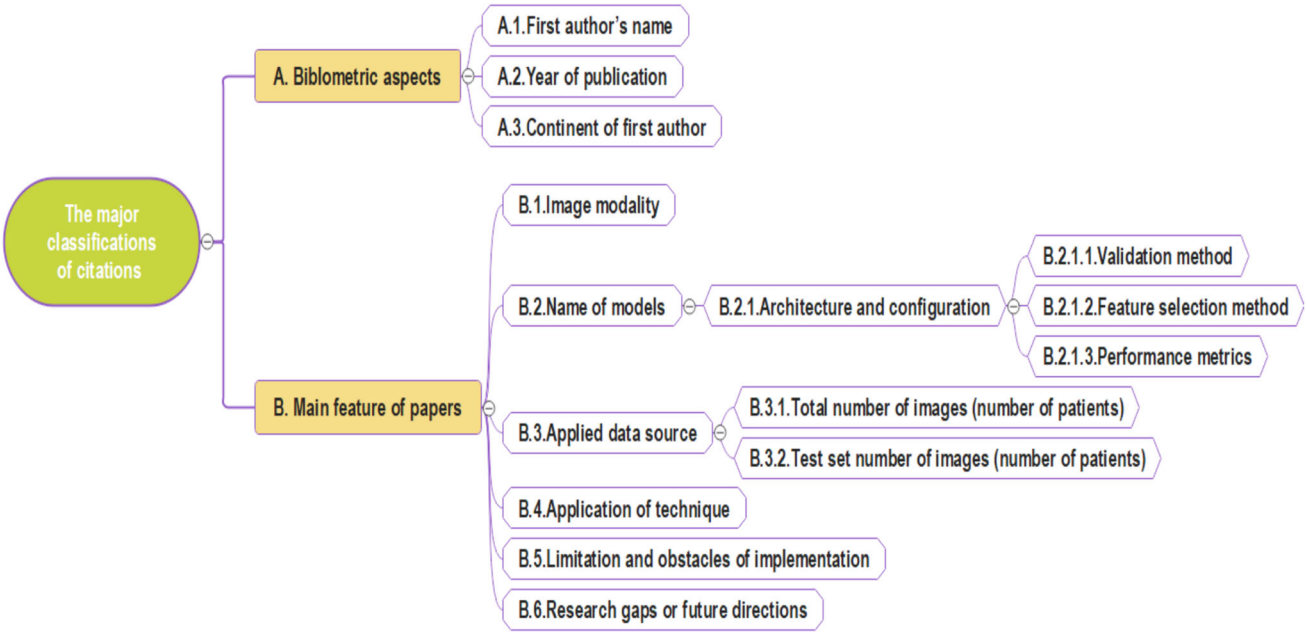

Fig. 1 Major extracted features and items of investigated papers

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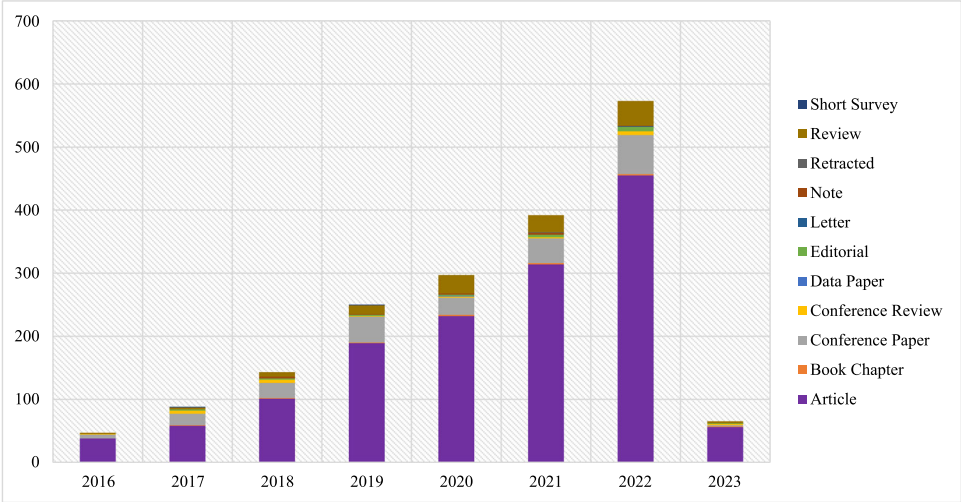

Fig. 2 Distribution of results based on the year and type of documents

3.1.2 Distribution of results in accordance with the subject area

The Scopus database provides "Subject areas" for search results that can identify researchers' interest in these specific topics. As shown in Fig. 3, the subjects are centered on medicine, computer science, biotechnology, engineering, and health professions.

3.1.3 Countries' bibliographic coupling

Using the VOS viewer, the number of articles published in each country was calculated, and international collaboration trends were studied. There were citations for 183 different countries and regions. The minimum number of documents in a country is adjusted to five; of 118 countries, 50 met the threshold. The size of each label in Fig. 4 indicates the number of citations. With 541, 516, and 187 articles, respectively, China, the USA, and India were the top three countries with a lot of documents. The connections between the nodes represented the nations' cooperative ties. The impact of a nation is depicted using linkages in Fig. 4. Importantly, the area of the purple circle that corresponds to the label "USA" is the largest, indicating that research carried out there has

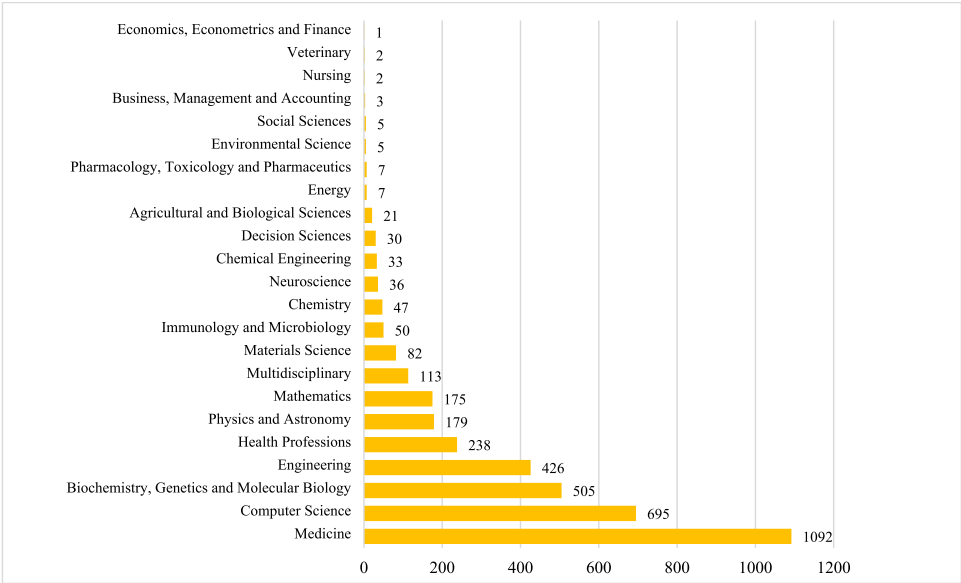

Fig. 3 Distribution of results by subject area

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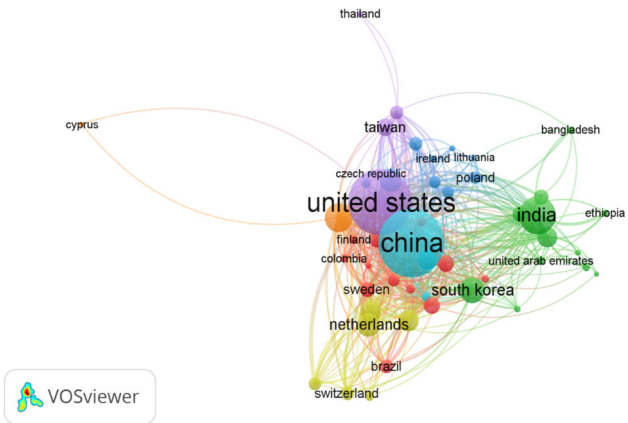
Fig. 4 Countries' network



had the greatest overall influence on the development of DL methods for breast cancer diagnosis based on photographs. The details related to the countries are given in Table 3. The USA and China are typically regarded as the countries with the most publications and the greatest impact. According to the analysis, USA and East Asian nations write most of the papers.

3.1.4 Scholars' co-authorship analysis

On the authorship level, scholars' co-authorship analysis was carried out using the full counting approach. The minimum number of papers for each author among the 6578 authors has been adjusted to 15. Based on the

Table 3 Detailed information on the 25 countries with the most publications

Ranking	Country	Number of documents	Citations	Total link strength
1	China	541	9564	238
2	USA	516	19,086	419
3	India	187	2572	94
4	UK	113	5299	204
5	South Korea	93	2974	84
6	The Netherlands	80	4943	128
7	Canada	75	2945	59
8	Germany	66	3489	134
9	Italy	66	1266	86
10	Saudi Arabia	62	1032	105
11	Japan	59	2067	52
12	Taiwan	48	640	54
13	Australia	44	1276	86
14	France	43	2962	80
15	Spain	43	2647	75
16	Pakistan	38	948	55
17	Sweden	37	1307	84
18	Egypt	35	728	34
19	Singapore	31	678	65
20	Brazil	29	1834	26
21	Iran	29	414	29
22	Switzerland	29	883	42
23	Austria	28	449	53
24	Portugal	26	1964	63
25	Turkey	26	1954	52

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number of documents and citations, the 35 most creative and significant authors were chosen. The results for the most creative and influential 32 scholars are shown in Table 4. Besides this, the network visualization of authors and cooperation is depicted in Fig. 5.

3.1.5 Analysis of keywords based on co-occurrence

The term-level analysis will provide several distinct clusters by highlighting the links between various concepts. It occurs when authors change terms in the article's title and abstract. The degree of relationship between the concepts depends on their distance from each other. A keyword must appear a minimum of five times in a sentence. In the abstract and title fields of the literature on the use of DL to make breast cancer diagnoses based on different photos, 183 keywords with a high degree of relatedness were discovered. Terms that are frequently used in relation to utilizing DL for the diagnosis of breast cancer are better categorized and assessed. In the abstract, keywords, or title of 1855 publications, the subject area of application of DL approaches for breast cancer diagnosis based on image processing was mentioned.

Table 4 Ranking the 20 most influential and productive researchers

Author	Documents	Citations	Total link
Yiming Wang	65	1162	64
Youru Li	40	405	51
Yanling Zhang	38	533	52
Jingyu Li	37	860	44
Yuan Liu	35	966	45
Sina Wang	33	464	57
Xuetao Wang	32	938	45
Jian Zhang	31	686	29
Xinmeng Li	29	497	27
Junxia Wang	28	452	39
Hong Li	25	462	26
Zijian Wang BS	23	254	35
Haicheng Zhang	21	413	20
Yang Chen	20	226	36
Li Wang	20	373	26
Lei Zhang	20	411	31
Xiaoling Zhang	20	340	29
Juan Chen	19	184	24
Lihua Li	19	415	27
Jingwen Liu	19	272	29
Hongyan Wang	19	142	19
Peter Chang	18	456	19
Chenchen Li	18	383	30
Jinjie Wu	18	722	20
Xin Yang	18	419	22
Hongde Liu	17	312	26
Bin Zheng	17	509	13
Houjin Chen	16	381	22
Richard Ha	16	296	12
Zhuangsheng Liu	16	261	32
Katja Pinker	16	391	0
Tianfu Wang	16	319	16

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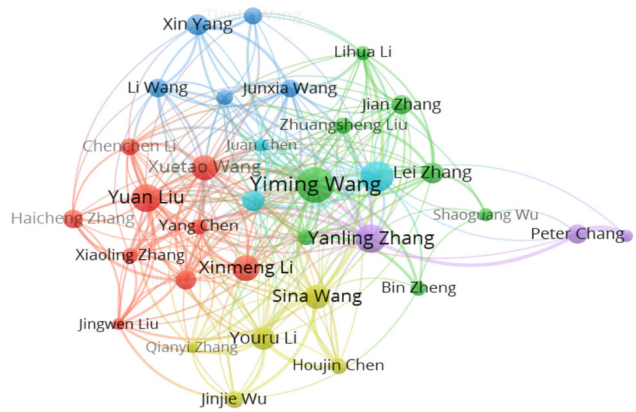

Fig. 5 Network visualization of 35 authors

Hence, based on word repetition, the keywords were found. The repeatability score for bibliographic descriptions was five. In total, 183 terms from the aforementioned criteria appeared at least five times among the 2753 words that were found. It was calculated using the binary counting method in the VOS viewer and the scores for each of the 183 terms. Figure 6 compiles the list of 183 pertinent words and displays the resulting map; it displays the nine clusters derived from the 183 words that appeared five or more times in the research article. Some of these 183 keywords occur regularly.

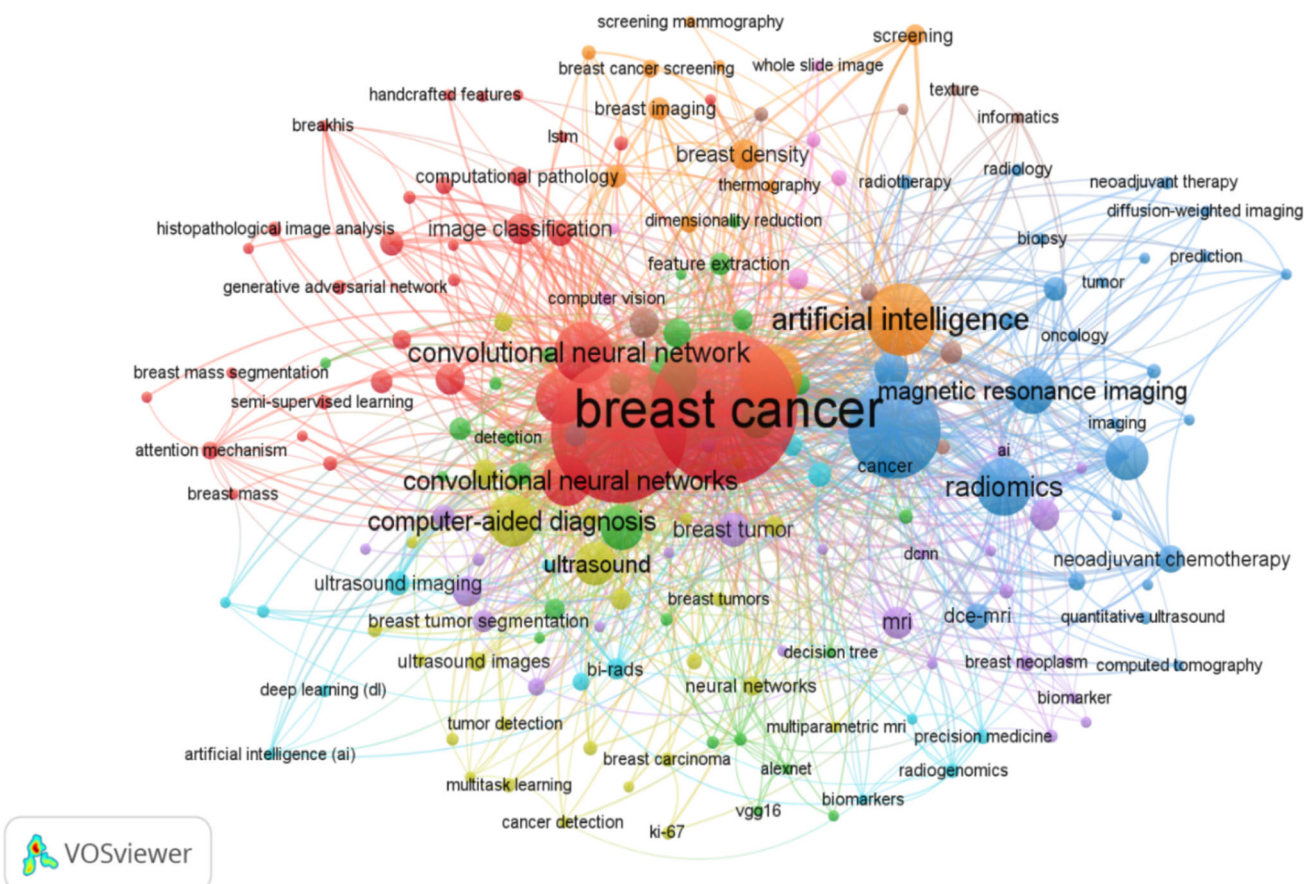

Fig. 6 Term map with colors indicating nine clusters

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3.2 Scoping review of papers

3.2.1 Study selection

Figure 7 shows the procedure for looking up papers using the PRISMA statement in scientific databases. A total of 2975 documents were received. In total, 1886 articles were left for screening after duplicates were eliminated. Another 1656 items were removed as a result of title and abstract screening. In the initial round, 230 articles were deemed pertinent, and their complete texts were examined. Finally, 135 papers were included in our systematic review after we read the article's full texts and used the inclusion and exclusion criteria. Table 5 and Table 6 in Appendix provide a summary of the major findings. Additionally, the architecture of methods and the main performance of applied techniques are described in Table 6 in Appendix A.

3.1.6 DL techniques

3.1.6.1 CNNs In neural networks, most deep architectures are built. ANNs are a type of information processing model created by copying biological neural networks, such as those found in the human brain [176]. This model's innovative information processing system structure, which comprises numerous units (neurons) with strong internal connections that cooperate to address certain issues, is its fundamental component. The neural network learns by transferring the knowledge or law concealed beneath the data to the network structure through the

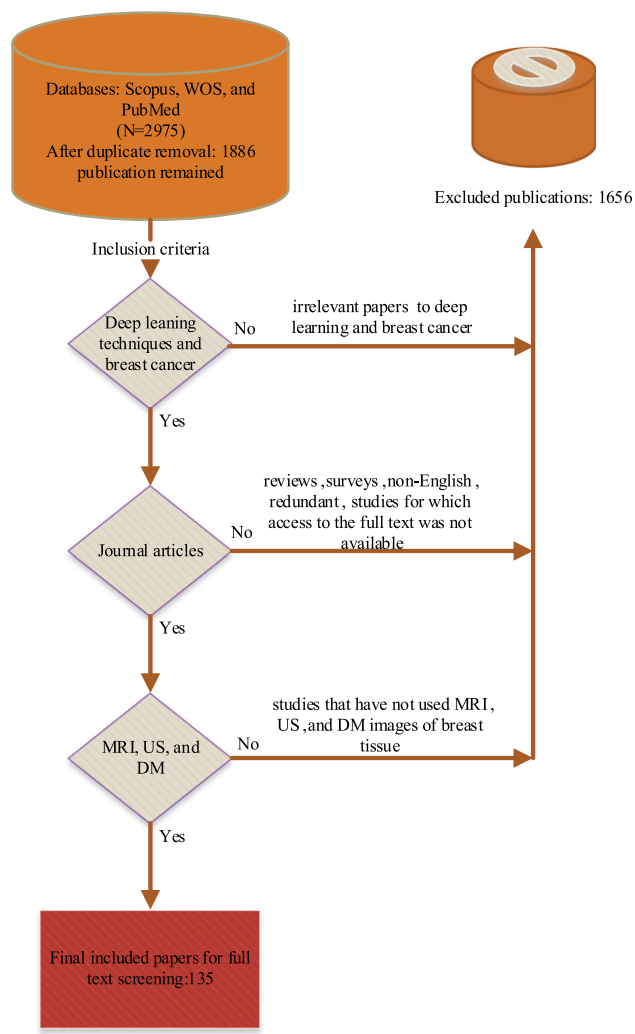

Fig. 7 Selection of the retrieved studies in given flow diagram

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Table 5 Highest performance metrics of models and developed architectures

References	Highest performance metrics						
	Accuracy (%)	Sensitivity (%)	Specificity (%)	AUC	Error	The Dice coefficient (%)	Jaccard index (IoU)
[44]	71	84	62	78	_	_	_
[45]	83	87	86	93	_	_	_
[46]	_	94	_	_	_	95	90
[47]	87	86	88	85	_	_	_
[48]	92	91	96	94	_	_	_
[49]	76	_	_	_	0.002	_	76
[50]	97	96	98	_	_	_	_
[51]	90	100	97	95	_	90	82
[52]	_	92	_	_	_	93	_
[53]	_	90	_	85	_	_	50
[54]	88	_	_	92	_	_	_
[55]	92	87	100	_	_	_	_
[56]	93	91	95	_	_	_	_
[57]	88	92	83	91	_	80	_
[58]	92	93	91	95	_	_	_
[59]	90	89	92	97	_	_	_
[60]	85	_	_	87	_	_	_
[61]	65	_	_	_	_	_	56
[62]	90	96	79	95	_	_	_
[63]	84	83	87	92	_	86	_
[64]	83	87	85	_	_	_	_
[65]	80	83	71	87	4	_	_
[66]	96	_	_	96	_	_	_
[67]	95	97	92	94	_	92	86
[68]	_	_	_	_	_	96	_
[69]	94	_	_	77	_	_	_
[70]	_	_	_	99	_	99	90
[71]	84	92	79	91	_	_	_
[72]	_	95	_	96	0.019	_	_
[73]	99	_	_	96	_	99	100
[74]	97	95	96	98	_	_	_
[75]	93	86	96	98	_	_	_
[76]	92	92	91	_	_	_	_
[77]	_	_	_	_	0.82	76	_
[78]	_	90	_	_	_	_	_
[79]	82	82	82	87	_	_	_
[80]	90	90	91	_	_	_	_
[81]	_	100	_	88	_	_	_
[82]	_	86	99	_	_	84	_
[83]	87	91	88	92	_	_	70
[84]	84	92	80	89	_	_	_
[85]	98	_	_	99	_	_	_
[86]	_	94	_	91	_	_	_
[87]	_	95	_	80	_	_	_
[88]	88	88	87	94	_	_	
[89]	97	91	96	_	_	_	_
[90]	_	_	_	69	_	_	_

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Table 5 (continued)

References	Highest performance metrics						
	Accuracy (%)	Sensitivity (%)	Specificity (%)	AUC	Error	The Dice coefficient (%)	Jaccard index (IoU)
[91]	_	_	_	_	_	87	78
[92]	99	_	_	_	_	82	73
[93]	92	97	89	95	_	_	_
[94]	98	_	_	95	_	_	_
[95]	98	_	_	99	_	_	_
[96]	80	_	_	80		94	87
[97]	_	_	_	_	_	77	64
[98]	99	100	94	96	_	_	_
[99]	96	97	_	95	_	_	_
[100]	_	_	_	_	_	74	64
[101]	82	_	__		_	82	95
[102]	87	60	95	85	_	_	_
[103]	82	85	81	90	_	_	_
[104]	92	93	92	94	_	_	_
[105]	_	_	_	98	_	91	_
[106]	92	95	88	97	_	_	_
[107]	85	89	82	92	_	_	
[108]	91	87	72	96	_	_	_
[109]	88	70	92	86			
[110]	93	94	91	_	_	_	_
[111]	_	_	_	73		_	_
[112]	_	91	78	87	_	_	_
[113]	99	_	_	_	_	_	98
[114]	81	_	_	89	_	_	_
[115]	99	_	_	_	_	_	_
[116]	_	_	_	82	_	_	_
[117]	96	97	97	98	_	_	_
[118]	92	98	100	97	_	_	_
[119]	89	**88**	**89**	93	_	_	_
[120]	98	95	96	_	_	_	_
[121]	85	_	_	94	_	_	_
[122]	97	_	_	_	_	_	_
[123]	_	_	_	_	_	97	94
[124]	_	_	_	_	_	_	91
[125]	80	_	_	93	_	_	_
[126]	98	92	98	_	_	91	85
[127]	78	66	84	74	_	_	_
[128]	81	90	75	_	_	_	_
[7]	_	_	_	90	_	_	
[129]	_	87	77	91	_	_	_
[130]	94	94	97	99	_	_	_
[131]	88	84	89	93	_	_	_
[132]	_	_	_	82	_	_	_
[133]	_	_	_	98	_	_	–
[134]	_	_	_	91	_	_	_
[8]	_	_	_	73	_	_	–
[135]	_	_	_	96	_	_	–

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Table 5 (continued)

References	Highest performance metrics						
	Accuracy (%)	Sensitivity (%)	Specificity (%)	AUC	Error	The Dice coefficient (%)	Jaccard index (IoU)
[136]	99	_	_	_	_	98	_
[137]	_	_	_	86	_	_	_
[138]	91	100	61	99	_	_	_
[139]	_	_	_	_	_	92	
[140]	75	49	92	71	–	_	
[141]	72	_	_	_	_	_	_
[142]	97	_	_	97	_	82	
[143]	92	90	93	93	_	_	_
[144]	96	96	98	_	_	_	_
[145]	_	_	_	_	_	82	_
[146]	95	97	_	95	_	_	_
[147]	88	_	_	_	_		
[24]	98	99	_	99	_	_	_
[148]	98	_	_	_	_	_	_
[149]	76	76	76	80	_	_	_
[150]	_	_	_	88	_	_	_
[151]	_	_	_	70	_	_	_
[152]	_	90	93	86	_	_	_
[153]	_	85	75	89	_	_	_
[154]	88	_	_	88	_	_	_
[155]	98	98	97	_	_	_	_
[156]	88	92	79	91		_	_
[157]	92	95	92	91	_	_	_
[158]	_	_	_	_	_	83	75
[159]	84	87	82	91	_	_	_
[160]	99	96	99	_	_	96	_
[161]	88	_	_	88	_		
[162]	_	_	_	96	_	_	_
[163]	95	98	90	_	52	_	_
[164]	96	95	96	_	_	_	_
[165]	90	99	83	87	_	_	_
[166]	94	99	99	97	_	_	_
[167]	90	_	_	85	_	22	84
[168]	85	80	93	92	_	_	_
[169]	98	98	99	98	_	97	_
[170]	_	97	_	_	_	_	_
[171]	_	_	_	93	_	_	_
[172]	91	91	92	_	_	_	_
[173]	_	_	_	_	_	78	_
[174]	83	91	70	85	_	_	_
[175]	–	71	80	82	32	_	_

processing of the experimental data. The most crucial aspect of an intelligent system is its learning capacity, and neural networks are frequently used in medical applications as a result [177].

An input layer, one or more hidden layers, and one or more output layers make up neural networks. A neural network receives an input, multiplies it by the network weights, and then outputs the result using a nonlinear

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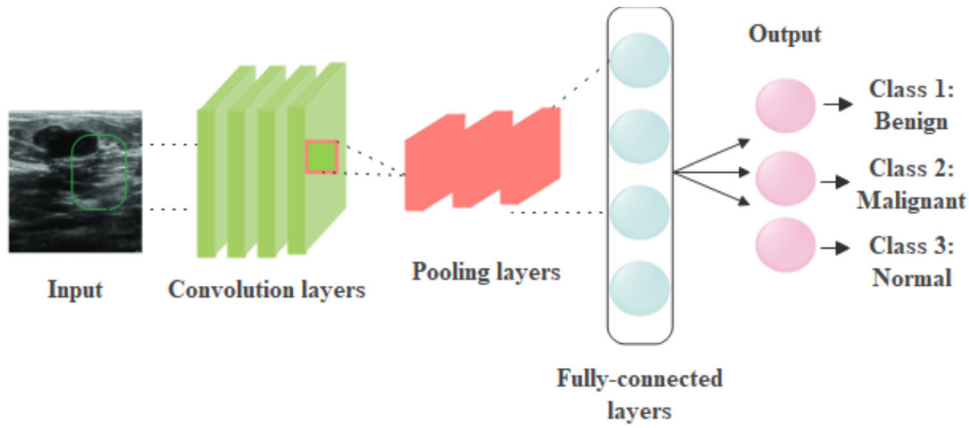

Fig. 8 Convolutional layers make up the majority of a CNN architecture, which also includes a pooling layer for feature map extraction with a function (SoftMax) and fully linked layers for classification. The output can be either benign, malignant, or normal classes

transformation function [178]. A group of neurons make up each hidden layer; each neuron may be completely coupled to every other neuron in the layer above. The categorization is carried out via the output layer, which is the final completely connected layer. DNNs are frequently referred to as such when they have numerous hidden layers [179].

DL includes different methods; one of the DL methods is CNNs, which have achieved significant success in image classification, especially image analysis (Fig. 8). Classification is the division of subjects based on measured characteristics into two or more classes, which can be used in the field of medical image analysis for disease identification and diagnosis [180]. The three main advantages of convolutional networks are [181]:

i. The weight-sharing mechanism (parameters) between different filters greatly reduces the number of parameters;
ii. Network connections are set locally and based on the two-dimensional structure of the input;
iii. Immutability and stability of the object's location change.

3.1.6.2 What is convolution? A function that is a modified version of one of the two original functions is produced by the mathematical operation of convolution, which operates on the two functions x and h. Under the name of "faltung," which translates as "folding" in English, convolution was first described by a German mathematician named Gustav Deutsch. The correlation function and convolution are comparable. This operator has uses in statistics, ML, image and signal processing, electrical engineering, differential equations, and other fields.

The convolution of two functions f and g is writ ten as the following relationship [182].

$$\mathbf{Z} = f * g \tag{1}$$

This relationship is defined as the integral of the product of two functions, the inverse of one of which slides over the other. By this definition, convolution is a special type of integral transform [183]:

$$f(t) * g(t) = \int_{-\infty}^{\infty} f(\tau)g(t-\tau)d\tau = \int_{-\infty}^{\infty} f(t-\tau)g(\tau)d\tau \tag{2}$$

The convolution formula can be considered as a weighted average of the function $f(\tau)$ with momentum t, where the weights are shifted (sliding) by $g(-\tau)$ by the amount t.

The discrete convolution of f and g' is obtained from the following equation [184]:

$$f[n] * g[n] = \sum_{m=-\infty}^{\infty} f[m]g[n-m] = \sum_{m=-\infty}^{\infty} f[n-m]g[m] \tag{3}$$

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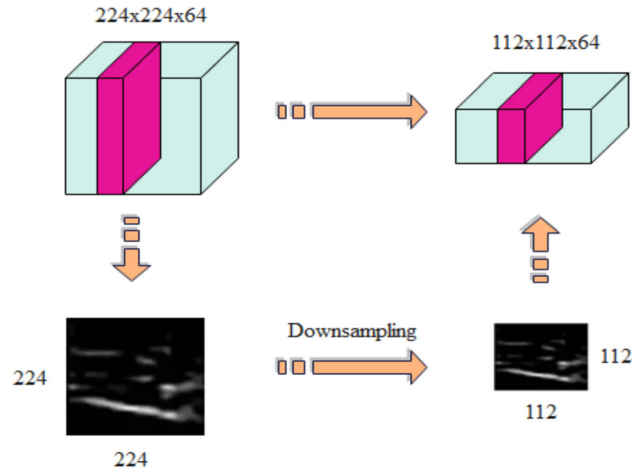

Fig. 9 Pooling layer on an image with dimensions 64 × 224 × 224 (computer science wiki)

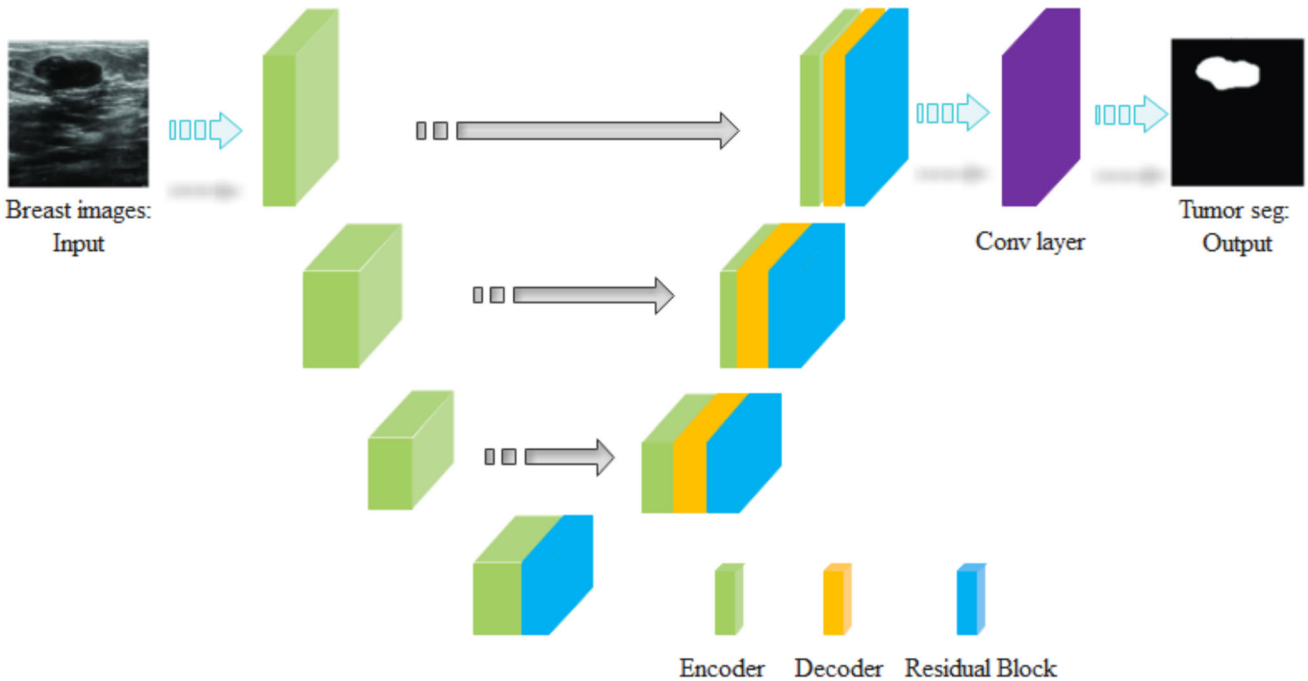

Fig. 10 An overall view of breast tumor segmentation

3.1.6.3 Pooling layers The main purpose of this layer is to reduce input dimensions and reduce sensitivity to unwanted changes in the image. This layer is usually placed after the convolution layer. Usually, in the convolution architecture, an integration layer is placed after several convolution layers. The features obtained by the convolution layer in the previous step can be classified using the usual classifiers. But this is very computationally heavy. The number of features extracted by the convolution layer from each of the subsections of the input is very large and it is likely to cause the system to not work properly. In fact, the integration layer is used to reduce the number of parameters (reduce the risk of overfitting), reduce the computational load in the network, and reduce memory. Since this layer calculates a constant function of the input, no parameters are added to the network. Each neuron in the integration layer, like the convolution layer, is connected to a limited number of neurons in the previous layer [30, 176, 185]. Neurons in this layer have no weight. In this layer, integration is usually done for each channel (depth) separately, so that the input depth and the output depth are equal. For this layer, two meta-parameters of perceptual area size and step size should be defined. There are different methods to implement this layer, the most common of which is the maximum integration function and the average integration function.

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In Fig. 9, the integration layer and how to reduce the dimensions are shown. As can be seen, the dimensions of the input mass are 224 × 224 × 64, which, after the integration operation, have been converted to 112 × 112 × 64. This operation only changes the length and width of the input, and the depth of the pile is kept unchanged.

3.1.6.4 Fully connected layer The fully connected layer or layers perform the operation of mapping two- or more-dimensional features to the one-dimensional output feature vector, and are placed after the last integration layer. The task of this layer is to calculate the final score for classification. This layer's output is a vector, with each element representing a score for a specific category. The output of this layer can be used to categorize or continue further processing. This layer is similar to the layers of artificial neural networks in the sense that each neuron in it is connected to all the neurons of the previous layer. It can be said that 90% of the parameters of a CNN network are in the fully connected layer.

Thus, the big problem with these layers is their large number of parameters, which causes a high processing cost during training. For this reason, one of the commonly used methods is to remove these layers or reduce the connections between these layers [186–188].

There are other ways to enhance a CNN's performance, including batch normalization and dropout. A regularization technique called dropout is employed to stop a CNN model from overfitting. CNN training is accelerated by a batch normalization layer, which also lessens the sensitivity to network initialization.

3.1.6.5 Famous pre-trained networks Deep CNNs require a lot of training data to avoid overfitting when training. Since in many fields, such as medicine, the amount of available data is limited to use a convolutional network and take advantage of it, pre-trained networks are usually used [189]. This means that a network trained on different data is used to analyze images in a new field. Some advanced pre-trained DL architectures are given in the below:

a. *AlexNet*: one of the pioneers of DL, Geoffrey Hinton, and his collaborators, introduced AlexNet as the first deep architecture. Without the amazing research made possible by AlexNet, which is a straightforward yet effective design, DL would not be what it is today [190].
b. *VGG:* the most distinctive feature of VGG is its pyramidal structure, where layers closer to the image are larger and layers farther away are deeper. Convolutional layers are the foundation of the VGG, and pooling layers are placed behind them. The layers are being reduced in size [191].
c. *ResNet*: one outstanding architecture that demonstrates the depth to which a DL architecture can go is ResNet. The basic component of the ResNet design, which stands for Residual Networks, is made up of a number of sedimentary modules that are stacked on top of one another [192].

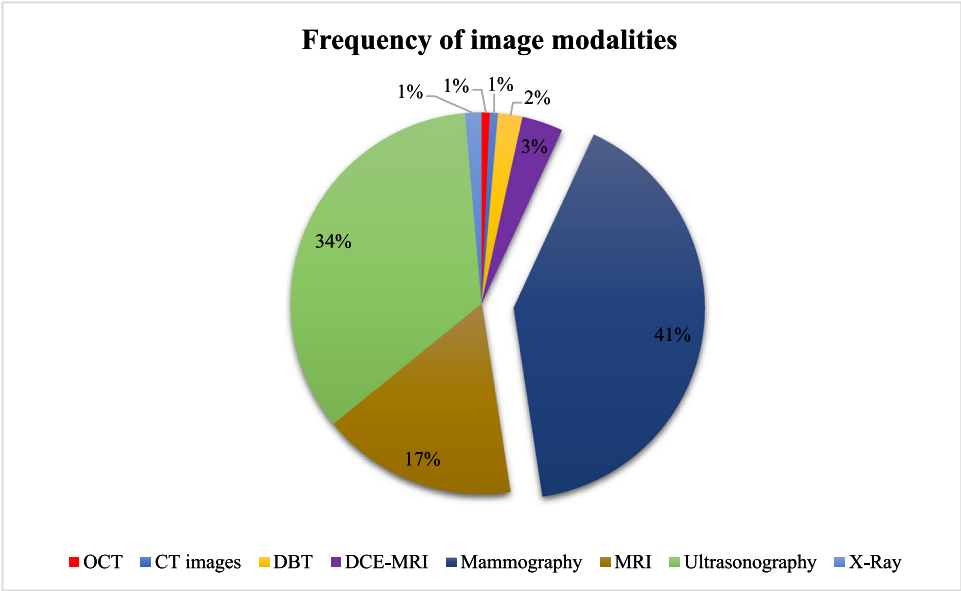

Fig. 11 A pie chart showing the modalities employed by various researchers

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d. *The Region-Based CNN* architecture is the most influential DL architecture ever used to solve object detection problems. The way this architecture works is that it tries to draw a box around all the objects in the image and then identify the object inside the image [193].
e. *YOLO*: YOLO is the most comprehensive real-time system for handling image recognition issues and DL. Before running the identification method in parallel for all of these parts to determine which category each component belongs to, the program first separates the image into various sections and labels each portion. It joins the things once they have been properly identified so that every pair of major objects is a box [194].
f. *SegNet* is a DL architecture used to solve image segmentation problems. This architecture consists of several processing layers (which do the same work as "encoders") and a series of opposite layers as "decoders" to specify the parts of the images [195].
g. *GAN*: GAN a completely other kind of neural network architecture called a GAN uses a neural network to create a brand-new image that is entirely absent from the training data yet sufficiently realistic to be included in this data set [196].
h. *U-Net*: the idea hidden in U-Net algorithm is to create a sequential shrinking path that happened by replacing the pooling layers instead of the sampling layers. In the following, after collecting the desired features, a sequential expansion path is used. The purpose of using such a path is to reconstruct the original image and spread the background information into it [197]. In short, it can be said that the U-Net network has two symmetric and identical expansion and contraction paths, so that whatever is removed in the contraction path is restored in the expansion path. The overall processing of breast tumor segmentation is depicted in Fig. 10.

3.1.7 Applied datasets

Digital breast cancer research databases, which are open to public or private clinical images, typically evaluate DL models. A growing number of medical photos are available online because most intelligent systems need a lot of data to function. To do this, DL algorithms are applied to existing digital public images from sources like the Curated Breast Imaging Subset of the DDSM Digital Database (CBIS-DDSM), the Mammographic Image Analysis Society (MIAS) database, Digital Database for Screening Mammography (DDSM), Automated Breast Ultrasound (ABUS) database, INbreast, Breast cancer screening–digital breast tomosynthesis (BCSDBT), Breast Cancer Digital Repository (BCDR), Dynamic contrast-enhanced magnetic resonance imaging (DCE-MRI) database, and Breast Ultrasound Image Dataset (BUSI), Digital Database for Breast Ultrasound Images (DDBUI).

In some studies, local databases have been used, whose images have been collected and tagged from medical institutions, hospitals, and universities. For instance, local datasets were collected from Jeonbuk National University Hospital (JNUH), Hunan Provincial Tumor Hospital (test cohort A), and Yunnan Cancer Hospital (test cohort B), two medical institutions: the Sun Yat-Sen University Cancer Center (SYSUCC) and Nanhai Affiliated Hospital of Southern Medical University (Foshan, China), Hospital of Shanxi Medical University, and Department of Galactophore Surgery and Department of Oncology of West China Hospital, Sichuan University. Employed datasets are given widely in Table 5.

3.1.8 Image modalities

In this review, only articles that used three main imaging modalities for training, testing, and validating deep networks were reviewed. These imaging types are US images, MRI, and X-ray images. The frequency of imaging modalities is given in Fig. 11. In some studies, several types of images were used for training and testing. Three of the imaging modalities were the most frequent, and a brief description of each is given below.

i. *Mammography*: digital mammography is the most used and important screening technique in clinical practice. It has the capacity to detect cancers before they intensify and become apparent or palpable to a physician. Digital mammography has drawbacks, such as being an inadequate screening method for

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women with large breasts since it includes ionizing radiation, despite the fact that X-ray mammography is the gold standard for breast cancer screening because it offers high 2D resolution. Digital mammography can therefore detect microcalcifications, which are exceedingly minute changes in utilized for network training and testing in 59 research (49%) overall.

ii. *Ultrasonography*: an ultrasound, which is operator-dependent and employs a handheld probe to target aberrant tissue around the breast, is susceptible to inaccuracy. Although a mammography is the best tool for spotting abnormalities deeper inside the breast tissue, an ultrasound is still highly helpful for assessing superficial tumors. Ultrasound method has also been widely used; this imaging modality was used in 50 studies (34%).

iii. *MRI*: besides from providing information on the vascularity of the breast tissue, MRI photographs the entire breast and delivers it as thin slices that span the complete breast volume. It has great potential for screening high-risk female patients and assessing treatment outcomes [40, 198]. This type of modality is only used in 24 of the publications referenced in this scoping review.

3.1.9 Performance metrics

The performance metrics used to evaluate DL are described in this section. If breast cancer is successfully diagnosed, it can be classified as true positive (TP) or true negative (TN), and if it is wrongly detected, it can be classified as false positive (FP) or false negative (FN). The most often used evaluation metrics for breast cancer classification and segmentation include accuracy, sensitivity, specificity, Dice similarity coefficient (DSC), intersection over union (IoU), area under the curve (AUC), and volume under the ROC surface.

3.1.9.1 Accuracy The proximity of a measured value to a reference or actual value is referred to as accuracy. In other terms, it refers to a tool's capacity to determine an exact value whose correctness can be assessed. Accuracy can be calculated using Eq. 4:

$$\boldsymbol{ACC} = \frac{TP + TN}{TP + TN + FP + FN} \tag{4}$$

Sensitivity This metric provides the proportion of correctly calculated positive instances among all positive cases. Put simply, it shows how accurately anticipated the total number of atypical breast cancer patients is. Equation 5 can be used to measure this:

$$\boldsymbol{Se} = \frac{TP}{TP + FN} \tag{5}$$

3.1.9.2 Specificity This metric shows how accurate the overall pessimistic forecasts are, while it indicates how many of the typical predicted is correct. Equation 6 can be used to indicate it [199]:

$$\boldsymbol{Sp} = \frac{TN}{TN + FP} \tag{6}$$

3.1.9.3 AUC Drawing receiver operating characteristics (ROC) plots can be helpful in presenting the findings for organizing and analyzing classifiers as well as showing their performance. ROC plots are frequently used in medical decision-making and have also lately come to light in DL and ML. Plotting the true positive rate (TPR) vs the false positive rate (FPR) for various threshold sets results in the ROC curve. Thus, the best results are when TPR is maximized and FPR is minimized. This indicates that the ideal point (FPR = 0 and TPR = 1) is in the upper left corner of the plot [200].

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3.1.9.4 DSC A statistical measure known as the DSC is used to evaluate the reproducibility of manual segmentation performance and the spatial overlap of anticipated image segmentation. Equation 7 can be applied to calculate this measurement [201]:

$$\boldsymbol{DSC}(\boldsymbol{A},\boldsymbol{B}) = \frac{2 \times |A \cap B|}{|A| + |B|} = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{7}$$

The ***IoU*,** which measures the overlap ratio between the target mask and the output of our prediction, is the area where the predicted segmentation and the actual segmentation overlap. This metric can be indicated by Eq. 8 [202]:

$$\boldsymbol{IoU}(\boldsymbol{A},\boldsymbol{B}) = \frac{A \cap B}{A \cup B} = \frac{TP}{TP + FP + FN} \tag{8}$$

3.1.10 Reported limitations and future directions

Despite the encouraging research's fruitful findings, there are still significant restrictions and difficulties in using DL algorithms for the identification and classification of breast cancer. Below is a discussion of the review's primary difficulties, underlying trends, suggested research directions, and challenges.

3.1.10.1 Reported limitations

1. The absence of extensive training datasets has been a significant barrier to developing DL models for medical imaging, as the authors of this study observed in the early stages of the studies they analyzed. Since the size and quality of the dataset have a substantial impact on how well the classifier performs, DL requires a large amount of training data. Lack of data is probably one of the main obstacles to DL's usefulness in medical imaging. Making vast amounts of medical imaging data is also challenging since, in order to eliminate human error, annotating the data takes a lot of time and work from a number of professionals as well as a person. Large medical imaging datasets are extremely challenging to build because, in order to eliminate human error, annotating the data takes a lot of time and effort from many experts as well as from a single person.
2. Because lesions may be obscured by dense tissues, such as dense breast tissue or intermediate mixed-type breast density, patients' chances of developing breast cancer may increase if AI-based approaches are used to detect lesions at an early stage.
3. The majority of the examined studies evaluated and assessed these using various datasets that were privately acquired by cancer research organizations or clinics. The efficiency of such algorithms across several studies is difficult to compare, which is one of this argument's main flaws. Employing data expansion techniques rather than transmitting learning to avoid overfitting is another drawback in some articles.
4. The researchers did not systematically assess how dense breast tissue impacts the capacity to detect breast abnormalities in mammography when using the DCNN model. High performance accuracy and longer execution durations can be traded off.

3.1.10.2 Future directions

1. In order to assess the generalizability of the model findings in a concealed or invisible collection of data, cross-validation is a technique for model validation. The goal is to categorize a dataset used to test a model during training in order to address issues with underfitting and overfitting as well as to demonstrate how the trained model generalizes to a different dataset.
2. In order to enable this dependency and incorporate data from several points of view, generic public image datasets with a variety of image modalities will be used. It will be profitable if researchers release a fall DNA

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case series. Additionally, it is advised to use 3D CNN for extracting 3D feature maps and multiple detection directions.

3. Images of every type of breast cancer case can also be gathered (iii). Because they allow researchers to assess the effectiveness of the recently developed multi-class breast cancer classification, boundary images should be labeled for the classification of multi-class breast cancer. The recommended strategy can be combined with various domain adaptation techniques that have been put out to deal with multi-domain issues in the researcher's future work.
4. The suggested methods will be gradually enhanced and validated by incorporating the advantages of end-to-end CNNs for clinical usefulness, following newly collected images, and fine-tuning model parameters.
5. Multimodality research needs to be looked into. Researchers have also investigated combining MRI and mammography imaging techniques. A fascinating extension will be to use mammography and MRI-based models as a foundation to analyze tomosynthesis images, which are gradually becoming a common breast imaging modality.
6. More advanced segmentation models, such as U-Net, have led to ground-breaking segmentation discoveries in various computer vision datasets. Moreover, applying this technique to several imaging modalities may enhance the results of breast cancer classification.

4 Discussion

Breast cancer diagnosis is one of the fields of interest for researchers. Medical images are one of the fastest, safest, and least invasive methods of diagnosis. Image analysis is a difficult and sometimes error-prone process. Therefore, employing faster and more accurate image analysis methods is essential. The early and accurate diagnosis of the tumor leads to the improvement of the treatment results and increases the life expectancy of the patients [203].

Between 2016 and February 2023, researchers discovered 1855 articles, of which 77.7% were original. In 2022, the number of articles has significantly increased. The increasing number of articles published regarding the use of these methods in the analysis of medical images demonstrates the high interest of researchers in this field (see Fig. 3). Most of the published articles revolved around the topics of medicine, computer science, biotechnology, engineering, and health professions. China, the USA, and India received the majority of the articles. Yiming Wang, a Chinese researcher, has written the most articles in this field. Most words repeated more than five times are "breast cancer," "convolutional neural network," and "artificial intelligence." Most studies use imaging techniques such as mammography, ultrasound, MRI, X-ray, DBT, CT scan, and OCT for the classification, screening, and origin of breast cancer. Additionally, researchers have applied mammography and ultrasound images more frequently than other images in most studies, indicating a greater willingness to use these two types of images in model development. Meanwhile, MRI is known as the best imaging technique for predicting tumor size in breast cancer. Histological type and tumor size are key parameters to consider when planning surgery [204].

Datasets such as DDSM, CBIS, INbreast, MIAS, BCDR, BUSI, and local datasets have been used in medical image classification models. Most studies have used the DDSM dataset (28 articles) and the local dataset (26 articles). The largest number of images is related to the study of Adam Yala et al., in 2019 which used 238,271 images from 66,661 patients [152]. The lowest number of images is related to the study of Michal Byra et al., which used 39 images from 30 patients [79]. Most of the researchers used more than 500 images.

Also, the findings of the research showed that most of the studies used the CNN model (75 articles) and the ResNet model (34 articles) to provide a model for the segmentation and classification of medical images. In addition, most studies have used U-Net to segment medical images. Each type of DL model has different applications [205].

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One of the usages of CNN is the analysis of images, especially the analysis of medical images for the purposes of classification, segmentation, and detection. One of the important advantages of CNN is the automatic extraction of image features, unlike traditional ML methods, which extract features manually [206].

The findings of this study showed that in all studies, models have been developed based on CNN. For tumor segmentation, almost all studies have used the U-Net model or applied changes in this model except for the study by Yongye Su and colleagues [100], which uses YOLO LOGO (Local–Global) architectures and the study by Pan Pan and his colleagues [77], which uses SC-FCN-BLSTM and the study of Rahima Rohvih, and Mehdi Jafari [51] which uses hybrid segmentation methods.

On the breast rim or chest wall, breast cancer lesions are typically not apparent. Breast ROI extraction is hence helpful for precise tumor segmentation. Medical image segmentation typically is done by the U-Net model. Based on the expert's designated breast region labels, a breast ROI drawing was produced using the U-Net model [207].

In most of the articles (73 articles), k-fold cross-validation has been used to evaluate the effectiveness of the classification models, of which 18 articles have used the fivefold cross-validation method. Other articles have not mentioned the evaluation of the effectiveness of the model. Ravi K. Samala and colleagues [134] used two-loop leave-one-case-out cross-validation to evaluate the effectiveness of the model, and also in the study of Agata Wijata [113], the validation frequency was 2 per epoch. The main application of this technique is to detect and classify malignant and benign lesions, screening, and segmentation. Most of the articles (77 articles) have used this technique to detect and classify malignant and benign lesions, and 7 articles have been used for screening.

Lately, DL techniques in the field of tumor detection have produced respectable results. The growing number of studies about the use of these techniques in medical imaging demonstrates the keen interest that researchers, particularly in affluent nations, have in this area [208].

The findings of this study also showed that most of the studies had been conducted in Asia (94 articles), especially in China. Also, 24 articles in this field have been published in Europe and 18 articles in USA. Some studies have shown that the interest in understanding and improving the adoption of AI techniques for educational purposes is greater than ever, not only in educational institutions but also in government sectors [208].

Today, many ML systems have been developed for commercial use. The very diverse applications of this technology are rapidly increasing its adoption in sectors such as healthcare, finance, and manufacturing [209]. Since DL technologies can dramatically increase productivity, they can be very important for the economic growth of China, which is facing an aging population. China is one of the global hubs for the development of AI and DL, with its largest technology companies making significant investments in research and development. Factors supporting the adoption of this technology in China are its large population and diverse industrial mix, which provide the potential to generate massive amounts of data and provide a huge market. The USA and European countries are also producing more impactful AI research with a strong ecosystem of AI startups [210].

One of the most important limitations raised in most studies is the lack of the size of the medical image dataset. It is challenging to use DL and exploit its capabilities on small amounts of data. DL software techniques must be trained on a large dataset to perform well. To overcome this challenge, it is possible to combine the data sets of different medical centers. Transfer learning or data augmentation can also be used to solve this problem [36].

Rapid advances in DL-based techniques in medical image analysis algorithms have enabled improved breast cancer diagnosis, treatment, and prognosis. Furthermore, the many capabilities and excellent performance of DL methods have attracted the attention of the research community. However, before DL can impact clinical practice, we need to address numerous research gaps. The following are the most significant research gaps identified in the studies: use of larger datasets, automatic ROI area extraction and lesion classification by the model, use of 3D CNN to extract 3D feature maps, use of strong preprocessing techniques to minimize unwanted effects, robustness development and validation of the proposed model, deployment of the model in a Web-based service, use of domain matching methods to deal with multi-domain problems, and the possibility of comprehensive analysis.

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4.1 Implications for practice

Health policymakers and decision-makers may gain from the use and advancement of DL-based radiomic techniques in intelligent systems and clinical decision support systems [211]. These variables might be taken into account for estimating the chance of getting breast cancer, projecting how the illness will progress, and evaluating how well therapies work. Nonetheless, further challenges have impeded the widespread application of AI methodologies in therapeutic settings. These problems draw attention to how important comprehensive validation, interpretability, and technology considerations are when putting DL solutions into practice [212].

The new claims have supplied crucial and current insights for radiologists who wish to apply AI in breast cancer care. These findings are based on a combination of the most recent advancements and trends with an analysis of basic concepts in DL techniques applied to medical imaging. Several studies have demonstrated that combining AI with less experienced radiologists can somewhat increase diagnostic accuracy. But unless AI is proven and integrated into practical tools, it would not have much of an impact on practice. DL-based systems that are both autonomous and supportive generally seemed to be slightly more accurate than the average human reader, while it is uncertain how sensitive they are. These models' ability to be developed into a ready-to-use tool that can assess whether the input data are adequate, evaluate the images, and provide the results to reporting radiologists is one of its strengths [213].

4.2 Limitations and Strengths of the study

Our study has both positive and negative aspects. The study's strengths are as follows:

- Have used a thorough search technique to locate a lot of research (1886 studies),
- Searching three significant databases to locate research,
- Six writers independently reviewing and analyzing studies to obtain data,
- Conducting thorough and universal bibliometric analysis.

In this investigation, we have run into limitations as well. DL in breast cancer detection is a rapidly growing field in the healthcare industry, and numerous research studies have been conducted in this area. As a result of our search of databases, a large number of relevant articles were identified, which made the researchers set a time limit. However, this study did not review any papers written in languages other than English, which might be a problem.

5 Conclusion

The current scoping review aims to investigate DL methods in breast cancer detection or segmentation. The application of DL methods in the last couple of years is promising because of the effectiveness of these methods in the early diagnosis of different cancers such as breast, lung, and skin. Based on the obtained results, it is difficult to comprehensively compare DL approaches with each other because of various factors. Some of these elements are: different image modalities; the sample size of images used for evaluation; and different assessment approaches such as training, testing, and validation methods.

Considering the techniques mentioned in this review, the CNN method has been widely used for breast tissue classification. A custom CNN design was suggested in some of the CNN-based investigations, while other studies used preexisting architectures—often by utilizing transfer learning—and yet other studies constructed entirely new architectures. It is not possible to determine the average performance value for all the investigations because the eligible articles for this review used a range of performance evaluation metrics. There are a number of restrictions that must be taken into account in order to determine the diagnostic accuracy of DL classifiers in clinical practice, including dataset sample size, image quality, and external technique validation. Furthermore, as

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a full database with the right attributes might not be available for breast cancer, a simulated training set might be a good alternative.

In future research, there are a few things that need to be taken into account: heterogeneity due to different sample sizes, unequal classes, and different CNN technique architectures that make it hard to say which method is best for a given task. As DL requires a huge number of annotated data points for modeling, it is recommended to establish a public image dataset with various image modalities for the specific case, such as breast cancer, to assist the dependency of different image modalities and integrate data from different dimensions.

Based on the findings, we deduced that AI-assisted imaging diagnosis presents a viable opportunity for a highly effective and precise breast cancer diagnostic model. Nevertheless, to obtain more optimization and validation, experimental research in clinical settings is required.

Large datasets must be gathered in order for academics to create precise models and algorithms that will serve as the central decision-making mechanism in decision-making systems in the future. We can expedite the process of developing an AI imaging platform for breast cancer diagnosis in the future by directly connecting AI-based diagnostic tools. These platforms need to be able to function properly, be transparent, have an understandable user interface, be self-optimizing, and be flexible enough to respond quickly to fresh data.

Appendix A

See Table 6.

Table 6 Summary of included papers

Contribution	Continents	Year of publication	Image modality	Model	Data source
Andrea Duggento et al. [44]	Europe	2019	Mammography	DCNN	CBIS-DDSM
Hongmin Cai et al. [45]	Asia	2019	Mammography	DCNN	The data sets were collected at two medical institutions: the Sun Yat-Sen University Cancer Center (SYSUCC) and Nanhai Affiliated Hospital of Southern Medical University (Foshan, China)
Han Jiao et al. [46]	Asia	2020	DCE-MRI	U-Net + + Faster RCNN	DCE-MRI Dataset
Xiaobo Lai et al. [47]	Asia	2020	DBT images	U-Net	DBT images database (collected at Zhejiang Chinese Medical University Affiliated Guangxing Hospital and Zhejiang Provincial Hospital of Traditional Chinese Medicine (TCM)
Saleem Z. Ramadan [48]	Asia	2020	Mammography	CNN	MIAS
Mohammed El Adoui et al. [49]	Asia	2019	DCE-MRI	Encoder and Decoder convolutional neural networks (CNN): SegNet U-Net	Breast MRI database (collected at Jules Bordet institute, Brussels, Belgium) + H8
Umar Albalawi et al. [50]	Asia	2020	Mammography	CNN	MIAS
Rahimeh Rouhia, Mehdi Jafari [51]	Asia	2015	Mammography	Hybrid segmentation methods	MIAS

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Table 6 (continued)

Contribution	Continents	Year of publication	Image modality	Model	Data source
Marco Caballo et al. [52]	Europe	2020	CT images	U-Net	Images were acquired by trained radiographers with a dedicated breast CT clinical system (Koning Corp., West Henrietta, NY)
Dezső Ribli et al. [53]	Europe	2018	Mammography	Faster RCNN	• DDSM • A dataset from the Semmelweis University in Budapest
Mohamed A. Hassanien et al. [54]	Asia	2022	Ultrasonography	ConvNeXt	BUS
Albin Sabani et al. [55]	Europe	2022	Mammography	DCNN	Local dataset
Shenbagavalli, Thangarajan [56]	Asia	2018	Mammography	Shearlet transform image enhancement method, neural network	DDSM
Hongwei Xue et al. [57]	Asia	2022	MRI	ADUNet	Breast MRI dataset collected from Tongji Hospital
Jun Bai et al. [58]	Asia	2022	Mammography	FFS-CNN, FFS-CNN-FC	• DDSM • CMMD • BCSDBT • A private data set provided by the Radiology department at the University of Connecticut Health Center (UCHC)
Haolin Yin et al. [59]	Asia	2022	MRI	CNN	A database of patients who underwent breast MRI examinations between March 2006 and May 2021 from PACS
Zilong Xu et al. [60]	Asia	2022	Ultrasonography	Three-block DenseNet-based deep learning model	A database of the Shandong Cancer Hospital and Institute
Yujie Li et al. [61]	Asia	2022	Ultrasonography	BUSnet model	BUS
Yanhong Chen et al. [62]	Asia	2022	DCE-MRI	Faster RCNN	Not mentioned
Jingjin Zhu et al. [63]	Asia	2022	MRI	VNet-based segmentation models, three ResNet-based characterization models (two single input and one multi-input models)	Data were chosen from the 1st Medical Center of PLA General Hospital database
Ying Zhou et al. [64]	Asia	2022	Ultrasonography	InceptionV3 DCNN	Data were chosen from the following three medical centers, including Zhejiang Cancer Hospital, Hebei Province Hospital of Traditional Chinese Medicine, and Yantai Affiliated Hospital of Binzhou Medical University
Chunxiao Li et al. [65]	Asia	2022	Ultrasonography	Multitask CNN model	Local dataset
Shivaji D. Pawar et al. [66]	Asia	2022	Mammography	DenseNet	DDSM
Mugahed A. Al-antari et al. [67]	Asia	2018	X-ray mammography	YOLO, FrCN, DCNN	INbreast
Gabriele Piantadosi et al. [68]	Europe	2019	MRI DCE-MRI	U-Net	• The first dataset is a private collection of labeled DCE-MRI images • A public dataset

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Table 6 (continued)

Contribution	Continents	Year of publication	Image modality	Model	Data source
Hua Li et al. [69]	Asia	2019	Mammography	AlexNet, VGGNet, GoogLeNet, DenseNet, DenseNet-Iineural	The dataset ate taken from mammography images provided by the First Hospital of Shanxi Medical University
Khaoula Belhaj Soulami et al. [70]	Asia	2021	Mammography	UNet	• DDSM database • CBIS-DDSM • Inbreast
Thomas Ren et al. [71]	Europe	2020	MRI	CNN	Data were obtained via a retrospective search of the Stony Brook Hospital
Woo Kyung Moon et al. [72]	Asia	2020	Ultrasonography	3D CNN: 3D simplified VGG16 and 3D DenseNet	3D ABUS
Mugahed A. Al-antari, Tae-Seong Kim [73]	Asia	2020	Mammography	YOLO CNN ResNet50 InceptionResNetV2	• CBIS-DDSM • INbreast
Manal AlGhamdi, Mohamed Abdel-Mottaleb [74]	Asia	2021	Mammography	DV-DCNN	• CBIS-DDSM • Inbreast
Xiaofeng Qi et al. [75]	Asia	2018	Ultrasonography	Mt-Net, Sn-Net	A database obtained from the Department of Galactophore Surgery and Department of Oncology of West China Hospital, Sichuan University
Jun Shi et al. [76]	Asia	2016	Ultrasonography	S-DPN	• A breast B-mode ultrasound dataset • A prostate ultrasound elastography dataset
Pan Pan et al. [77]	Asia	2020	Ultrasonography	SC-FCN-BLSTM	ABUS dataset
Ming Fan et al. [78]	Asia	2019	DBT	Faster RCNN	A dataset collected at Fudan University affiliated Cancer Center
Michal Byra et al. [79]	Europe	2020	Ultrasonography	InceptionResNetV2 CNNs	ImageNet dataset
Neeru Singla et al. [80]	Asia	2019	OCT images	InceptionV3	Local dataset
Natalia Antropova et al. [81]	Europe	2019	DCE-MRI	VGGNet LSTM	DCE-MRI Dataset
Yuehang Wang et al. [82]	Asia	2020	Mammography	AM-MSP-cGAN	• CBIS-DDSM, • INbreast
Feiqian Wang et al. [83]	Asia	2020	Ultrasonography	DCNN 3D U-Net	ABUS dataset
Bo-Yang Zhou et al. [84]	Asia	2021	Ultrasonography	ACNN models (The monomodal ACNN model, The dual-modal ACNN model, The multimodal ACNN model)	Local dataset
Hiba Chougrad et al. [85]	Asia	2018	Mammography	VGG16, ResNet50 Inception v3	• DDS, • INbreast, • BCDR

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Table 6 (continued)

Contribution	Continents	Year of publication	Image modality	Model	Data source
Ravi K. Samala et al. [86]	Asia	2018	Mammography	DCNN	A database was collected in the Department of Radiology at the University of Michigan Health System (UM)
Tsung-Chen Chiang et al. [87]	Asia	2019	Ultrasonography	3D CNN	ABUS dataset
Yi Wang et al. [88]	Asia	2020	Ultrasonography	CNN	Dataset from Jeonbuk National University Hospital (JNUH)
João Otávio Bandeira Diniz et al. [89]	America	2017	Mammography	CNN	DDSM
Ewan Evain et al. [90]	Europe	2021	Ultrasonography	Mask-RCNN	Images were collected from several French institutions
Haonan Yang, Dapeng Yang [91]	Asia	2022	Ultrasonography	CSwin-Pnet	UDIAT Diagnostic Center of the Parc TaulíCorporation, Sabadell (Spain) and Baheya Hospital, Cairo (Egypt)
Se Woon Cho et al. [92]	Asia	2022	Ultrasonography	BTEC-Net RFS-UNet	• BUSI • UDIAT dataset
Michal Byra. [93]	Europe	2021	Ultrasonography	ResNet101	BUSI
Meng Jiang et al. [94]	Asia	2020	Ultrasonography	ResNet50	Two independent test sets were obtained from Hunan Provincial Tumor Hospital (test cohort A) and Yunnan Cancer Hospital (test cohort B)
Meteb M. Altaf. [95]	Asia	2021	Mammography	A hybrid model based on pulse-coupled neural network (PCNN)–DCNN	• DDMS, • INbreast, • BCDR datasets for training and testing • MIAS for testing only
Vivek Kumar Singh et al. [96]	Asia	2018	Mammography	cGAN A multi-class CNN	• Inbreast, • DDSM, • Hospital Sant Joan de Reus dataset
Xuyang Cao et al. [97]	Asia	2020	Ultrasonography	Auto-DenseUNet based on the neural architecture search (NAS)	• ABUS • 3D cardiac MRI dataset
Mohammed A. Al-masni et al. [98]	Asia	2018	Mammography	YOLO FC-NNs	DDSM
Jing Niu et al. [99]	Asia	2021	Mammography	CBAM	• DDSM • Inbreast
Yongye Su et al. [100]	Asia	2022	Mammography	YOLO LOGO (Local–Global) architectures	• CBIS- DDSM • INBreast
Richard Ha et al. [101]	America	2018	MRI	3D CNN	An institutional PACS database
Richard Ha et al. [102]	America	2019	MRI	CNN	Data were collected in retrospective study
Heqing Zhang et al. [103]	Asia	2020	Ultrasonography	VGG16 VGG19 ResNet50	Data were collected in a retrospective study
Akiyoshi Hizukuri et al. [104]	Asia	2020	MRI	A CNN derived with Bayesian optimization: AlexNet, ZFNet, VGG16, GoogLeNet	Data were collected in a cohort study

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Table 6 (continued)

Contribution	Continents	Year of publication	Image modality	Model	Data source
Yunzhi Huang et al. [105]	Asia	2019	Ultrasonography	Grading network (G-CNN) ROI identification network (ROI-CNN)	Local dataset
He Ma et al. [106]	Asia	2021	Ultrasonography	Fus2Net	Local dataset
Yang Gu et al. [107]	Asia	2022	Ultrasonography	VGG19	Data was collected from multicenter cohorts (30 hospitals)
Jaeil Kim et al. [108]	Asia	2021	Ultrasonography	Weakly supervised DL algorithms: VGG16, ResNet34, GoogLeNet	Local dataset
Hamidreza Taleghamar et al. [109]	America	2022	Ultrasonography	Residual network and residual attention network (RAN)	Data was collected from unnybrook Health Sciences Centre (SHSC), Toronto Canada
S. Phani Praveen et al. [110]	Asia	2022	Mammography	ResNet	The dataset is open-source Invasive Ductal Carcinoma (IDC) that are part of Breast Histopathology Image
Xiaofei Zhang et al. [111]	Asia	2018	Mammography	CNN	Local dataset from University of Kentucky
Tomoyuki Fujioka [112]	Asia	2020	Ultrasonography	Xception, InceptionV3, InceptionResNetV2, DenseNet121, DenseNet169, NASNetMobile	Local dataset
Agata Wijata [113]	Europe	2021	Ultrasonography	CNN	Local dataset
Helen ML Frazer et al. [114]	Europe	2021	Mammography	ResNet	BreastScreen Victoria dataset
Moi Hoon Yap [115]	Europe	2016	Ultrasonography	Patch-based LeNet, U-Net, FCN-AlexNet	Local dataset
John Arevalo [116]	south america	2015	Mammography	CNN	BCDR
Mahmoud Ragab [117]	Asia	2022	Ultrasonography	VGG, ResNet	Benchmark Breast Ultrasound Dataset
alexanDer Ciritsis [118]	Europe	2018	Mammography	CNN	A local image archiving and communications system at Zurich
Ting Xiao [119]	Asia	2018	Ultrasonography	InceptionV3, ResNet50, Xception	A local hospital-based data (Sun Yatsen University)
N. Ashokkumar [120]	Asia	22,020	Ultrasonography	CNN	A local data base, Erasmus Medical Center
Bashir Zeimarani [121]	South America	2020	Ultrasonography	CNN	A data set from Federal University of Rio de Janeiro – Brazil
Chia-Hung Lin [122]	Asia	2022	Mammography	CNN	MIAS
Andrik Rampun [123]	Europe	2020	Mammography	CNN	• MIAS, • INBreast, • BCDR, • CBIS-DDSM
Mohammad Alkhaleefah [124]	Asia	2022	Mammography	SegNet	• INbreast • CBIS-DDSM, • One private dataset obtained from Cheng Hsin General Hospital in Taiwan

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Table 6 (continued)

Contribution	Continents	Year of publication	Image modality	Model	Data source
Naveed Chouhan [125]	Asia	2021	Mammography	Deep Convolutional Neural Network (DFeBCD Technique)	Dataset related to Image Retrieval for Medical Application (IRMA)
ChuanBo Qin [126]	Asia	2022	MRI	U-Net	Local dataset
Li Li [127]	Asia	2022	MRI	CNN	Local dataset
Xuehua Xiao [128]	Asia	2022	Ultrasonography	CNN	Local dataset
Natalia Antropova [7]	North America	2017	MR Mammography Ultrasonography	CNN	Local dataset
Ayelet Akserlod-Ballin et al. [129]	Europe	2019	Mammography	Combined machine- and deep learning model: DNN Gradient boosting algorithm (GBM)	Data were chosen from cohort study
Yeşim Eroğlu et al.[130]	Europe	2021	Ultrasonography	Hybrid-based CNN system: Alexnet MobilenetV2 Resnet50 SV, KNN	The Breast Ultrasound Data Set at Behaye hospital
Michal Byra et al. [131]	Europe	2019	Ultrasonography	VGG19 SVM	Data set at the Moores Cancer Center, University of California,San Diego
Ravi K Samala et al. [132]	America	2018	Mammography	ImageNet LSVRC-2010 VGG, ZFnet, GoogLeNet Microsoft ResNet	Dataset was collected at The Institutional Review Board
Songfeng Li et al. [133]	Asia	2019	Mammography	DCNN	Dataset was collected at The Institutional Review Board
Ravi K. Samala et al. [134]	America	2020	DBT images Mammography	ImageNet-trained AlexNet CNN	Local dataset from the University of Michigan DDSM
Dooman Arefan et al. [8]	America	2021	Mammography	GoogLeNet LDA Classifier (denoted as GoogLeNet-LDA)	Data were chosen from cohort study
Sarah S. Aboutalib et al. [135]	America	2019	Mammography	AlexNet	• FFDM • DDSM
Yang Zhang et al. [136]	Asia	2020	MRI	U-Net	Data were chosen from retrospective study
Nan Wu et al. [137]	America	2019	Mammography	ResNet Auxiliary Patch-Level Classification BI-RADS Classification	Local dataset
Jiejie Zhou et al. [138]	Asia	2020	MRI	ResNet50 CNN	Data were chosen from retrospective study
Arun Asokan Nair et al. [139]	America	2021	Ultrasonography	U-Net VGG13 BatchNorm MaxPool	Local dataset
Simukayi Mutasa et al. [140]	America	2021	Mammography	DCNN	Data were chosen from retrospective study
Richard Ha et al. [141]	America	2021	Mammography	CNN	Data were chosen from retrospective study
Michal Byra et al. [142]	Europe	2021	Ultrasonography	U-Net SK-U-Net	Data were chosen from retrospective study
Yang Zhang et al. [143]	Asia	2021	MRI	CNN CLSTM	Data were chosen from retrospective study

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Table 6 (continued)

Contribution	Continents	Year of publication	Image modality	Model	Data source
Siyuan Lu et al. [144]	Asia	2022	Ultrasonography	ResNet18 SAFNet ELM RVFL	USI dataset can 130 be downloaded from Kaggle
Viksit Kumar et al. [145]	Asia	2018	Ultrasonography	U-Net	Local dataset
Chen Zhang [146]	Asia	2020	Mammography	Multi-view feature fusion neural network model (MVNN) DenseNet MLO CSAM	Advanced Biomedical Research
Maleika Heenaye-Mamode Khan et al. [147]	Asia	2021	Mammography X-rays MRI	CNN ResNet50	• CBIS-DDSM • UPMC Network Cancer Registry Database
Tariq Mahmood et al. [24]	Asia	2022	Mammography	ConvNet VGGNet, InceptionV3, ResNet152V2, InceptionResNetV2	MIAS
Feng-Zhou Zhang et al. [148]	Asia	2022,	Mammography	MCNN MPN	MIAS
Meng Zhu et al. [149]	Asia	2022	Ultrasonography	InceptionV3, ResNet50, ResNet101, DenseNet161, DenseNet169,	DCIS (213 with MIC) in West China Hospital of Sichuan University,Sichuan Provincial People's Hospital
Daniel Truhn et al. [150]	Europe	2019	MRI	DCNN	Data was collected from cohort study and Local institutional review board approval was obtained
Adam Yala et al. [151]	Europe	2019	Mammography	ResNet18 CNN	IBIS
Adam Yala et al. [152]	Europe	2019	Mammography	ResNet18 CNN	Data extracted from retrospective study, tumor registries of five hospitals (academic and general)
Li-Qiang Zhou et al. [153]	Asia	2020	Ultrasonography	Inception V3, InceptionResNetV2, ResNet101	Tongji Hospital Hubei Cancer Hospital
Ridhi Arora et al. [154]	Asia	2019	Mammography	CNN AlexNet, VGG16, ResNet, GoogLeNet,InceptionResNet	CBIS-DDSM
Revathy Sivanandan, Jayakumari. [155]	Asia	2021	Ultrasonography	CNN StepNet	Local dataset
Mohammed El Adoui et al. [156]	Europe	2020	MRI	Multi-input deep learning DL networks	Local dataset
Tomoyuki Fujioka et al. [157]	Asia	2019	Ultrasound	GoogLeNet	Local dataset
Haibo Huang et al. [158]	Asia	2021	Ultrasonography	Deep Residual U-Net MResUNet model	Local dataset
Huiling Xiang et al. [159]	Asia	2021	Ultrasonography	3D Res-CapsNet convolutional neural network	ABUS
Lu Huo et al. [160]	Asia	2021	MR images	nnU-Net	Local dataset
Thomas Ren et al. [161]	America	2021	MRI	CNN 2D DCE	Dataset was collected from the Stony Brook Hospital
Jan Witowski et al. [162]	America	2022	MRI	CNN	Dataset was collected the New York University (NYU) Langone Health breast imaging sites

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Table 6 (continued)

Contribution	Continents	Year of publication	Image modality	Model	Data source
Fei Gao et al. [163]	America	2018	Mammography MRI CEDM	SD-CNN GBT ResNet50	Dataset acquired from tertiary medical center (Mayo Clinic Arizona), and a dataset acquired from the hospital housed at Zhejiang University, China
Mohammad I. Daoud et al. [164]	Asia	2020	Ultrasonography	CNN VGG19	BUS
Khalil ur Rehman et al. [165]	Asia	2021,	Mammography	Computer vision-based FC-DSCNN VGGNet-16–19 AlexNet Resnet50	Data were chosen from retrospective study from Local (PINUM) hospital in Pakistan,Diagnostic Imaging Nuclear Medicine and Radiology Research
Kuen-Jang Tsai et al. [166]	Asia	2022	Mammography	DNN EfficientNet SENet	Dataset was collected E-Da hospital, Taiwan
Wei-Chung Shia et al. [167]	Asia	2022	Ultrasonography	DeepLab v3 + ResNet50 Xception Mobilenet V2	Data were chosen from retrospective study
Haolin Yin et al. [168]	Asia	2022	MRI	CNN ResNet ImageNet	Data collected from retrospective study
Kuochen Zhou et al. [169]	Asia	2022	Mammography	Deeplab v3 +	Mini-MIAS, Inbreast
Yi Wang et al. [170]	Asia	2019	Ultrasonography	3D CNN U-Net DDS ReLU TM BN	BUS
Xin Shu et al. [171]	Asia	2019	Mammography	CNN RGP GGP image net	CBIS-DDMS
Manal AlGhamdi et al. [172]	Asia	2017	Mammography	DU-Net CNN DenseNet	CBIS-DDSM
Masoomeh Rahimpour et al. [173]	Europe	2022	MRI	3D U-Net	Data collected from cohort study
Juan Zhou et al. [174]	Asia	2019	MRI	Deep 3D densely connected networks:CNN DenseNet	Data collected from cohort study
Yuchen Qiu et al. [175]	Asia	2017	Mammography	CNN MLP	Local dataset

Contribution	Total number of images/masses/ examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Andrea Duggento et al. [44]	1158 images (Not mentioned)	Fourfold cross-validation	1) Screening mammography 2) Detect and classify malignant and benign lesions	Not mentioned	1) Applying a larger dataset
Hongmin Cai et al. [45]	990 images (Not mentioned)	Tenfold cross-validation	1) Microcalcification discrimination for breast cancer screening	(1) Limitation of data in dataset	1) The region of interest (ROI), region extraction, and lesion classification to be automatically implemented by the CNN method Should be considered in future works
Han Jiao et al. [46]	Not mentioned (75 Patients)	Fivefold cross-validation	1) Breast masses segmentation 2) Detect and classify malignant and benign lesions	(1) Limitation of data in dataset	1) Establishment of larger open data and the opinions of radiologists and surgeons in clinical practice 2) Detection from multiple directions as well as using 3D CNN for extracting 3D feature map

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Table 6 continued

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Xiaobo Lai et al. [47]	87 DBT volumes (71 patients)	Fivefold cross-validation	Breast masses segmentation	Lack of sufficient DBT image data	1) Combination of 3D reconstruction image information and 2D efficient data information
Saleem Z. Ramadan [48]	322 images (Not mentioned)	Not mentioned	Detect and classify malignant and benign lesions	Limitation of available labeled training data	1) Applying a larger dataset
Mohammed El Adoui et al. [49]	86 images (43 patients)	Not mentioned	1) Breast masses segmentation	1) Due to the lack of data, only 2D slices were used as inputs rather than 3D volumes 2) Only one collaborator radiologist was working to provide the ground truth labels	1) Improve the robustness of the proposed model 2) Use the segmented tumor method as a part of an end-to-end breast cancer response to a chemotherapy prediction system 3) Deploy the proposed model to a Web-based service 4) Explore and concept a cloud resource
Umar Albalawi et al. [50]	322 images (Not mentioned)	Not mentioned	1) Detect and classify malignant and benign lesions	Not mentioned	1) The changing of dataset based on the disease in training and testing
Rahimeh Rouhia, Mehdi Jafari [51]	57 images (Not mentioned)	Tenfold cross-validation	1) Breast masses segmentation 2) Detect and classify malignant and benign lesions	1) The presence of noise or artifacts in high levels in the MIAS database	1) Apply strong preprocessing techniques to minimize the unwanted effects of such intruders in future works
Marco Caballo et al. [52]	69 images (Not mentioned)	Not mentioned	1) Breast mass segmentation 2) Detect and classify malignant and benign lesions	1) Limitation of size of the dataset	1) In future work, with an expanded dataset, this study will be included in the development of an automated computer-aided diagnosis system for dedicated breast CT images
Dezső Ribli et al. [53]	* DDSM: 2620 images (NOT mentioned), * Semmelweis University database: 847 FFDM images (174 patients)	Not mentioned	1) Detect and classify malignant and benign lesions 2) Localize malignant lesions	1) Limitation of size of the dataset	1) The expensive, traditional CAD solutions, which have controversial efficiency, could be replaced with the recently developed, deep learning-based, open-source object detection methods in the near future
Mohamed A. Hassanien et al. [54]	9156 images (59 patients)	Not mentioned	1) Predicting Breast Tumor Malignancy	Not mentioned	1) Validating the proposed method in another breast. 2) Improve the quality of BUS images, and improve the classification rate 3) The researchers will consider using ROIs instead of the entire BUS image to further enhance the classification results

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Table 6 continued

Contribution	Total number of images/masses/ examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Albin Sabani et al. [55]	1744 images (438 patients)	Tenfold cross-validation	1) Classification of Mammographic Soft Tissue Opacities 2) Decision-making in the presence of Breast Cancer 3) Highlight benign and suspicious soft tissue opacities in the mammograms	1) Limitation of data was used. 2) The researchers cannot exclude that different machine learning algorithms other than dCNNs might reach a higher accuracy on the available amount of data 3) Only four different dCNN configurations have been tested 4) The researchers did not systematically evaluate how dense breast tissue influences the detectability of breast lesions in the mammography using their dCNN model 5) The researchers also did not evaluate how the presence of additional features such as microcalcifications might influence the performance of their DCNN model	1) The proposed technique might be used as a standardized quality control tool, providing the expertise of a team of radiologists
Shenbagavalli, Thangarajan [56]	300 images (Not mentioned)	Not mentioned	1) Detect and classify malignant and benign lesions	Not mentioned	1) In future the possibilities of considering the most optimal set of features are to be done in order to create more accurate classification
Hongwei Xue et al. [57]	435 images (87 patients)	Not mentioned	1) Breast masses segmentation	1) All data in this paper are from the same domain and independent identically distributed	1) Many domain adaptation methods are proposed to tackle multi-domain issues, and they can be applied to the proposed approach in the researcher's future work
Jun Bai et al. [58]	13,290 images (Not mentioned)	Not mentioned	1) Detect and classify malignant and benign lesions	1) Limitations in the structure of the backbone network due to the computational complexity	1) The researchers will work on developing models to segment tumors using current year and prior year images as their future work
Haolin Yin et al. [59]	700 images (700 Patients)	Not mentioned	1) The preoperative differentiation of Breast ductal carcinoma in situ and invasive breast cancer	1) The limited number of patients 2) Selection bias may be present in the study 3) The blocks were not the best choice for sampling peritumoral information	1) Applying a larger dataset

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Table 6 (continued)

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Zilong Xu et al. [60]	Not mentioned (144 Patients)	Not mentioned	1) Predict breast cancer molecular subtypes, predict the expression of Human epidermal growth factor receptor 2 (HER2)	1) The ultrasound images to build the model were collected from only one manufacture (Philips) 2) Only one type of ultrasound image was used to build the model	1) The feature concatenation of convolutional operation in the neural network should be explored to build a two-branch model
Yujie Li et al. [61]	780 images (Not mentioned)	Not mentioned	1) Detect and classify malignant and benign lesions	1) Limitation of the dataset size	1) Extending the capacity of the labeled dataset
Yanhong Chen et al. [62]	364 images (364 patients)	Fivefold cross-validation	1) Predicting malignancy among the breast lesions	1) Specificity of the model was relatively low when distinguishing breast lesions rather than images 2) Only DEC-MR images used in this study 3) Because of the architecture of the CNN, the internal parameters or weights of each layer were invisible	1) Future research may incorporate additional risk factors, such as clinical data, such as age, family history of breast cancer, menopausal status, and multi-parametric MRI characteristics 2) Although the precise cause is yet unknown, this type of inaccurate classification may be the result of the CNN's restricted capabilities, which will be improved in future research
Jingjin Zhu et al. [63]	Not mentioned (5811 patients)	Not mentioned	1) Breast masses segmentation 2) Lesion Characterization	1) It is difficult to determine from the presented results how the developed models might perform with data obtained under different protocols because only data from two medical centers were used 2) The researchers are unable to determine whether the multi-input model will significantly improve the diagnostic performance of radiologists	1) Collecting representative, large, and multi-institutional datasets to test the CADx model
Ying Zhou et al. [64]	707 images (707 patients)	Not mentioned	1) Detect and classify malignant and benign lesions 2) Differentiating non-lactating mastitis and malignant breast tumors	1) There is not much data in this study 2) The AI automatic classification system lacks sufficient data for in-depth analysis, and the outcomes may be biased and inaccurate	1) Actively building open databases 2) Optimizing the features of small data sets (fine annotation) on the basis of big data 3) Developing multimodal ultrasound AI

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Table 6 (continued)

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Chunxiao Li et al. [65]	5911 images (2131 patients)	Not mentioned	1) Prediction of lymph node metastasis 2) Detect and classify malignant and benign lesions	1) Limited size of dataset 2) Data were only derived from two-dimensional US images	1) Increased data expansion using larger and multicenter datasets 2) To enhance the diagnostic performance of ALN status, future studies should concentrate on a combination of multi-mode US images of breast masses
Shivaji D. Pawar et al. [66]	800 images (Not mentioned)	Fivefold cross-validation	1) Classify mammographic breast density (MBD) as per the breast imaging reporting and data system (BI-RADS) classification	1) Less image data is used in this investigation, and no image enhancement techniques are applied to increase the dataset 2) Only one sort of dataset is addressed in the proposed study	1) Data enhancement techniques 2) Training the model with different vendor-specific image datasets and testing results of all the mentioned state-of-the-art methods with the proposed method with the same work environment
Mugahed A. Al-antari et al. [67]	410 Images (115 patients)	Fourfold cross-validation	1) Detect and classify malignant and benign lesions 2) Breast tumor segmentation or localization	Not mentioned	1) Applying a larger dataset
Gabriele Piantadosi et al. [68]	130 images (109 patients)	Tenfold cross-validation	1) Breast tissue segmentation, segment breast parenchyma from the air and other tissues	1) The limitation may be represented by the spatial resolution of the images: the approach fails when it is trained on a dataset with a smaller resolution and then applied on a dataset with a larger resolution	1) Applying images with high resolution
Hua Li et al. [69]	30,630 images (2042 patients)	Tenfold cross-validation	1) Detect and classify malignant and benign lesions	1) The limitation of the training samples 2) The limitations of shallow learning	1) Applying a larger dataset
Khaoula Belhaj Soulami et al. [70]	1079 images (Not mentioned)	Not mentioned	1) Breast tumor segmentation or localization 2) Detect and classify lesions into normal, benign, and malignant	1) There is no preprocessing step before the training procedure in the strategy	1) By employing data augmentation, cross-validation, fine-tuning more model parameters, increasing the number of training epochs, and using transfer learning in the future, the performance of the proposed model can be enhanced

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Table 6 (continued)

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Thomas Ren et al. [71]	259 images (99 patients)	Fivefold cross-validation	1) Prediction of axillary lymph node metastasis	1) Small sample size 2) Proposed algorithm was trained with one image type 3) Proposed algorithm is not fully automated	1) Train the CNN model using multi-parametric MRI data types to improve prediction accuracy
Woo Kyung Moon et al. [72]	333 images (246 images)	Not mentioned	1) Breast masses segmentation	Not mentioned	1) The suggested system will be steadily enhanced and verified by incorporating the advantages of end-to-end CNNs for clinical utility, fine-tuning model parameters, and monitoring fresh collected ABUS images
Mugahed A. Al-antari, Tae-Seong Kim [73]	DDSM: 10,480 images (2620 patients) Inbreast: 360 images (115 patients)	Fivefold cross-validation	1) Detect and classify malignant and benign lesions	1) Limitation of the size of training and testing annotated breast lesion images	1) Applying a larger dataset
Manal AlGhamdi, Mohamed Abdel-Mottaleb [74]	410 images (Not mentioned)	Not mentioned	1) Detect and classify malignant and benign lesions	1) The restriction of the analysis to mammograms	1) These models were demonstrated to have produced greater outcomes when the number of training samples was low by using a tensor-based learning approach 2) A comprehensive mammography analysis framework that is able to recognize, match, and classify various breast masses
Xiaofeng Qi et al. [75]	8145 images (2047 patients)	Not mentioned	1) Detect and classify malignant and benign lesions	Not mentioned	1) Assessing if ultrasound screening is useful for directing clinical operations using deep neural networks 2) Increasing the dataset's size while using it in real-world scenarios 3) Integrating many image modalities, which could make the diagnosis more precise
Jun Shi et al. [76]	200 images (Not mentioned)	Fourfold cross-validation	1) Detect and classify malignant and benign lesions	Not mentioned	1) The proposed S-DPN DL algorithm suggests that S-DPN has the potential for other medical images

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Table 6 (continued)

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Pan Pan et al. [77]	170 images (124 patients)	Not mentioned	1) Breast masses segmentation	1) Nipple that appeared in consecutive slices had similar echo texture with tumor and falsely detected (the proposed method segmented the tumor and nipple as whole)	1) Automatically locating and removing the nipple region using the orthogonality view
Ming Fan et al. [78]	105 images (89 patients)	Not mentioned	1) Detect and classify malignant and benign lesions	1) Data limitations for training	1) Using a large training and testing dataset
Michal Byra et al. [79]	39 images (30 patients)	Not mentioned	1) Early prediction of breast cancer response to neoadjuvant chemotherapy	1) Patient enrollment restrictions	1) Raw radio frequency US data might be used to construct NAC response prediction models
Neeru Singla et al. [80]	Not mentioned (48 patients)	Not mentioned	1) Prediction of healthy and malignancy breast tissue, 2) Assessment of breast cancer margin, localize the region of breast cancer	Not mentioned	1) By automatically extracting concealed image information, the suggested method will be useful in the future for monitoring individual patient health progress and therapeutic success
Natalia Antropova et al. [81]	703 images (Not mentioned)	Not mentioned	1) Detect and classifiy malignant and benign lesions	1) Inadequate dataset availability	1) Using a larger dataset
Yuehang Wang et al. [82]	CBIS-DDSM: 795 images (Not mentioned), Inbreast: 107 images (Not mentioned)	Not mentioned	1) Breast masses segmentation	Not mentioned	1) Provide a new end-to-end architecture that unifies segmentation and classification processes and commits to ROI-approximate segmentation and classification outputs
Feiqian Wang et al. [83]	Not mentioned (263 patients)	Fivefold cross-validation	1) Detect and classifiy malignant and benign lesions 2) Breast masses segmentation	1) The size of the dataset was small 2) There was some variation in the percentage of benign and malignant nodules 3) A few lesions were omitted because to non-nodular breast disease and invisible or difficult-to-detect ABUS	1) Using a more comprehensive dataset

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Table 6 (continued)

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Bo-Yang Zhou et al. [84]	818 images (807 patients)	Not mentioned	1) Preoperative prediction of molecular subtypes, predicting triple-negative from non-triple-negative breast cancers	1) IHC phenotypes, not genetic subtypes, were used to classify breast cancer molecular subtypes 2) This research only assessed multimodal Ultrasound images; other imaging modalities (such as mammography and MRI) were not examined 3) The validation set sample size in this investigation was small	1) Future research should investigate how to decode the molecular subtypes of breast cancer shown on multimodal US imaging using the constantly updated AI algorithms
Hiba Chougrad et al. [85]	6116 images (1529 patients)	Fivefold cross-validation	1) Detect and classify malignant and benign lesions	Not mentioned	1) To overcome the obstacle posed by mammograms of exceptionally thick breasts, deeper structures and more difficult images must be employed
Ravi K. Samala et al. [86]	2282 images (Not mentioned)	Fourfold cross-validation	1) Detect and classify malignant and benign lesions	1) The DBT test set is too small 2) A prototype system, whose shape differs from clinical systems, was used to obtain the DBT	1) Using a bigger dataset
Tsung-Chen Chiang et al. [87]	230 images (187 patients)	Fivefold cross-validation	1) Detect and classify malignant and benign lesions	1) Tumor boxes localization must be improved 2) The system does not draw the shape of the discovered tumor	1) Study of segmentation approaches such as traditional image analysis techniques (watershed, active contours, etc.) or, more recently, fully convolutional networks (FCN) for pixel-wise dense predictions in semantic segmentation
Yi Wang et al. [88]	316 images (263 patients)	Fivefold cross-validation	1) Detect and classifiy malignant and benign lesions	Not mentioned	1) To assess current work using specialized detection algorithms for breast lesion identification and classification as an end-to-end CAD solution

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Table 6 (continued)

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
João Otávio Bandeira Diniz et al. [89]	2482 images (Not mentioned)	Not mentioned	1) Detect and classify malignant and benign lesions 2) Breast masses segmentation	Few pixels reach the fully linked layer due to the several layers of convolution and pooling As a result, the network finds it challenging to distinguish	1) Using the CNNs' architecture to enhance the outcomes of the phases where they are applied; 2) Using a large number of empirical factors 3) Developing new methods (such as the use of evolutionary algorithms)
Ewan Evain et al. [90]	460 images (Not mentioned)	Fivefold cross-validation	1) Detect and classify malignant and benign lesions	1) The percentage of images in each BI-RADS class in the training and test datasets may also have an impact on the outcomes 2) The ultrasound data only included breast nodules that were previously classified as BI-RADS 3 or 4	1) The Mask-RCNN option includes a detection threshold below which outputs are not considered 2) Concentrate more specifically on the segmentation phase of the Mask-RCNN to determine the form properties of the nodules
Haonan Yang, Dapeng Yang [91]	1) 163 ultrasound images (Not mentioned) 2) 780 ultrasound images (Not mentioned)	Fivefold cross-validation	1) Breast masses segmentation	Not mentioned	1) Investigating more effective feature extraction modules. These include incorporating CNNs into transformers or developing more powerful transformer variations to improve perception and extraction of breast lesion locations and boundary information
Se Woon Cho et al. [92]	1) 780 images (Not mentioned), 2) 163 images (Not mentioned)	Fivefold cross-validation	1) Detect and classify the stage of tumor 2) Breast masses segmentation	Not mentioned	1) Creating a technique that can do classification and segmentation in the same network 2) Using deep learning-based augmentation approaches to protect the data needed for network training
Michal Byra. [93]	647 images (Not mentioned)	Not mentioned	1) Detect and classify malignant and benign lesions	Not mentioned	1) 1) It is planned to study additional possible forms for the DRS layers 2) Test the method's utility utilizing other network architectures

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Table 6 (continued)

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Meng Jiang et al. [94]	4828 images (1275 patients)	Not mentioned	1) Breast cancer subtype classification: triple-negative, HER2 + , luminal A, and luminal B	1) Establishing a causal relationship between visual representations and particular molecular patterns is difficult 2) The annotated data in this investigation were based on IHC surrogates rather than formal genetic analysis, which may have resulted in bias 3) A limited number of individuals with HER2 + and triple-negative tumors were studied. 4) Tumors that are luminal B/HER2-positive and luminal B/HER2-negative have distinct imaging characteristics and therapy methods	1) A clinical study based on genetic analysis is needed in the future to examine the clinical diagnostic validity and determine whether the DCNN model's categorization of molecular subtypes agrees with formal genetic analysis 2) Enroll more patients who fit these subtypes so that the DCNN model may be trained and used more effectively 3) A more accurate model that forecasts the two types independently and is based on a bigger sample size
Meteb M. Altaf. [95]	INbreast: 410 images (Not mentioned), BCDR: 736 lesions (344 patients), DDMS: 900 images (450 patients)	Tenfold cross-validation	1) Classification of benign and malignant: Normal, Abnormal Benign, and Abnormal Malignancy	Not mentioned	1) Expanding the combined dataset by examining other datasets. The authors' last concern is making sure the model avoids to fall in overfitting
Vivek Kumar Singh et al. [96]	Inbreast: 410 images (Not mentioned) DDSM: 1;168 cases of breast tumors Local dataset: 300 malignant tumors	Tenfold cross-validation	1) Classify the generated masks into four tumor shapes: irregular, lobular, oval and round and Segmentation of ROI	1) In order to center the tumor within a loose frame crop and get the highest accuracy, previous knowledge about the tumor's position must be supplied 2) The cGAN model is set up to segment tumors that are completely contained inside the ROI; otherwise, the model fails to segment it	1) Developing a multistage framework to identify additional breast tumor characteristics (such as margin type and microcalcifications) that will be included into a more thorough diagnostic to determine the breast cancers' level of malignancy

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Table 6 (continued)

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Xuyang Cao et al. [97]	1) 170 volumes (from 107 patients): 120 volumes (from 75 patients) for training and other 50 volumes (from 32 patients) for testing 2) 3D cardiac MRI dataset has 20 volumes for training and other 10 volumes for testing with large variability	Fivefold cross-validation	1) Breast masses segmentation	1) Auto-DenseUNet makes the assumption that the dataset utilized in the searching and training operations will have the same data distribution; nevertheless, image distributions in various hospitals with varying collection equipment and settings may differ significantly 2) Even if the two datasets are from the same imaging modality, the performance of the best design may suffer when applied to a different dataset	1) Based on the successful experience of U-Net, a search space with a larger degree of freedom will be constructed to further enhance the suggested approach
Mohammed A. Al-masni et al. [98]	3000 images (Not mentioned)	Fivefold cross-validation	1) Detect and classify malignant and benign lesions	Not mentioned	1) The proposed CAD system will next be evaluated in practice to determine its true validity
Jing Niu et al. [99]	1647 masses (Not mentioned)	Tenfold cross-validation	1) Detect and classify malignant and benign lesions	1) This research does not classify calcification lesions 2) Digital breast tomosynthesis (DBT) technology is progressively replacing FFDM in clinical diagnosis, however there is no DBT image database that is accessible to the general public	1) Study the characteristics of calcification lesions and suggest a network model that may be used to diagnose them 2) Collaborate with the hospital to perform DBT imaging study 3) Provide a comprehensive automated mammography image-assisted diagnostic system so that scientists may continue to develop techniques for identifying, classifying, and diagnosing breast lesions

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Table 6 (continued)

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Yongye Su et al. [100]	1699 images (Not mentioned)	Not mentioned	1) Detect and classify malignant and benign lesions 2) Breast masses segmentation	1) The suggested model is not fully trained. Both the gated axial transformer plus LOGO training strategy-based segmentation model and the YoloV5L6-based breast mass detection model are trained individually 2) Only CBIS-DDSM data were used to train the models; the INBreast dataset served as an extra test set to see how broadly applicable the model was	1) Making the system end-to-end by combining the YoloV5L6 and LOGO loss functions into one 2) Employing several imaging modalities to lessen performance bias
Richard Ha et al. [101]	1114 images (137 patients)	Fivefold cross-validation	1) Quantification of breast MRI fibroglandular tissue (FGT) and background parenchymal enhancement (BPE)	1) A limited feasibility assessment at one institution	1) Test on new datasets gathered from other sources 2) Include preprocessing for bias field correction, a denoising technique, and data normalization for proposed algorithm's input 3) To choose the optimal hyperparameters, random search or the Bayesian search approach would be used 4) A larger dataset will probably further enhance the model, which might eventually be used in the clinic to predict the probability of breast cancer in patients
Richard Ha et al. [102]	(216 patients)	Not mentioned	1) Predict the molecular subtype of a breast cancer: luminal A, luminal B, HER2, and basal	1) Limited sample size 2) Absence of external evaluation poses 3) Genetic analysis is used to define different molecular subtypes, which is a technically and economically difficult procedure 4) Utilizing IHC surrogate instead of genetic analysis to determine molecular subtype	1) A CNN's performance will increase as data sets grow 2) Making use of prospective validation studies and external validation datasets

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Table 6 (continued)

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Heqing Zhang et al. [103]	5000 images (Not mentioned)	Fivefold cross-validation	1) Detect and classify malignant and benign lesions	1) The professional labeling of the ultrasound images used in this study made it wasteful and difficult to conduct future large-scale data research 2) The absence of consistent image acquisition standards may have had some influence on the study's findings 3) Information such as the quantity of benign and cancerous images, the specific pathological type, the mass size, and other details were unknown	1) Professional labeling of the images was effective and helpful for upcoming large-scale data investigation
Akiyoshi Hizukuri et al. [104]	56 DCE-MRI examinations—each of which contained five sequential phase images (56 patients)	Threefold cross-validation	1) Detect and classify malignant and benign lesions	1) A 2D-based DCNN model with a limited dataset was used 2) The 3D-based DCNN model needs a lot of training data since it has many parameters to learn	1) A bigger dataset must be used in the subsequent research to test the strategy for optimizing the hyperparameters and the suggested DCNN model 2) A 3D-based DCNN model may be more suitable than a 2D one
Yunzhi Huang et al. [105]	531 cases of Category 3, 443 cases of Category 4A, 376 cases of Category 4B, 565 cases of Category 4C, and 323 cases of Category 5	Not mentioned	1) Tumor identification and the tumor grading: Category "3," Category "4A,"Category "4B," Category "4C," and Category "5"	Not mentioned	1) Gathering additional information, especially on "Category 4B" and "Category 5," to improve the proposed grading system's prediction accuracy 2) Adding the images from categories 0, 1, 2, and 6 to the present grading system
He Ma et al. [106]	1052 images (Not mentioned)	Tenfold cross-validation	1) Detect and classify malignant and benign lesions	1) There are significant disparities between publicly available datasets of natural sceneries and medical data	1) Collect experimental data and keep refining the model by taking transfer learning into account

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Table 6 (continued)

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Yang Gu et al. [107]	14,043 images (5012 patients)	Not mentioned	1) Detect and classify malignant and benign lesions	1) Only B-mode US images were used to train the DL model 2) Getting images from the same device in each of the 30 hospitals	1) Adding clinical data, elastography, and molecular data to the model's training set 2) Data gathered from diverse devices made by various vendors might help the DL model perform better, and this is an important area for future research
Jaeil Kim et al. [108]	1000 images (Not mentioned)	Not mentioned	1) Detect and classify malignant and benign lesions	1) Using smaller image sizes 2) Due to the intricacy of the data curation procedure, time- and labor efficiency were not explicitly assessed	1) Making use of datasets with different institutions, imagers, and US devices
Hamidreza Taleghamar et al. [109]	Not mentioned (181 patients)	Not mentioned	1) Predicting breast tumor response to neoadjuvant chemotherapy (NAC)	1) The available dataset for training and improving the models was rather small	1) The availability of bigger datasets in the future would allow for more thorough model training and a more thorough assessment of their effectiveness in making predictions for new patients
S. Phani Praveen et al. [110]	198,738 IDC negative and 78,786 IDC positive	Not mentioned	1) Detect and classify of ductal carcinoma	1) The screening procedure and clinical results community	1) Finding the anomaly in cancer's early stages
Xiaofei Zhang et al. [111]	3290 images (Not mentioned)	Not mentioned	1) Detect and classify malignant and benign lesions	1) Utilizing negative frames in a positive test might deceive the model's training	1) To attain the best performance, create an assembled classifier that combines the data from 2D mammography and 3D tomosynthesis
Tomoyuki Fujioka [112]	304 images (294 patients)	Fivefold cross-validation	1) Detect and classify malignant and benign lesions	1) Not all masses were identified by cytological or histological procedures 2) Many radiologists conducted ultrasonography SWE exams, which may have generated an image bias	1) To confirm results, substantial multicenter prospective research is needed, employing a variety of ultrasonic devices
Agata Wijata [113]	619 images (91 patient)	The validation frequency was 2 per epoch	1) Automatic Biopsy Needle Detection	1) If CNNs are not correctly trained, phantom datasets, which do not reflect the real-world artifacts present in clinical data, may induce overfitting	1) Use modified parameters to overcome overfitting the CNN

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Table 6 (continued)

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Helen ML Frazer et al. [114]	28,694 images (7498 cases)	Not mentioned	1) Detecting the presence of breast cancer	1) Unless the same data are utilized for training and testing, it is difficult to draw conclusions from comparisons of AUC and ACC values	1) Further research is required regarding the ethical, legal, and societal ramifications of integrating AI into healthcare systems
Moi Hoon Yap [115]	469 images (Not mentioned)	Tenfold cross-validation	1) Detect and classify malignant and benign lesions	1) The training procedure, which takes time and requires a representative collection of typical images, is the biggest downside	1) The goal of future research is to improve accuracy by including additional training data, to expand on existing work to include breast ultrasonography lesion segmentation and classification, and to assess how well the whole CAD framework performs
John Arevalo [116]	736 image (344 patients)	Not mentioned	1) Detect and classify malignant and benign lesions	Not mentioned	1) To improve the representation, future study will use bigger structures and various image modalities
Mahmoud Ragab [117]	780 images (Not mentioned)	Not mentioned	1) Detect and classify malignant and benign lesions	Not mentioned	1) Deep instance segmentation methods may be created in the future to increase the EDLCDSBCDC technique's detection rate
alexanDer Ciritsis [118]	20,578 image (5221 patient)	Fourfold cross-validation	1) Detect and classify malignant and benign lesions	1) These models offer an overall proportion of breast tissue, which may not represent high-density parenchymal patterns in specific parts of the breast	1) The integration of dCNN into clinical workflow may contribute to enhancing the diagnostic validity and dependability of mammographic breast density evaluation in the clinical routine
Ting Xiao [119]	2058 image (Not mentioned)	Tenfold cross-validation	1) Detect and classify malignant and benign lesions	Not mentioned	1) Using a bigger dataset
N. Ashokkumar [120]	908 images (Not mentioned)	Not mentioned	1) Predicting the axillary lymph node metastasis	1) It is unclear how such an algorithm might examine incoming data, create links with prediction labels, and do such analysis	1) To accurately predict primary breast cancer in the future, a logistic regression model using the ANN model will be utilized
Bashir Zeimarani [121]	641 images (Not mentioned)	Fivefold cross-validation	1) Detect and classify malignant and benign lesions	1) To avoid the overfitting issue, a minimal number of hidden layers were utilized in this study's dataset, which was quite small	1) More datasets will be gathered and various CNN architectures with additional hidden layers will be used in further research

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Table 6 (continued)

Contribution	Total number of images/masses/ examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Chia-Hung Lin [122]	322 images (Not mentioned)	Tenfold cross-validations	1) Breast Lesions Screening	1) AI-based approaches may not effectively detect lesions at the early stage because they may be shadowed by dence tissues, such as dense breast tissue or intermediate mixed-type breast density, increasing patients' chance of developing breast cancer	1) It is proposed that the automatic screening model could take the place of the conventional CNN methods for certain tasks requiring knowledge and experience in medical image examinations, such as diagnostic mammogram, CT, and MRI. This would lessen the workload and allow doctors to concentrate on follow-up decision-making and medical strategies
Andrik Rampun [123]	1087 images (Not mentioned)	Cross-validation	1) Breast Pectoral Muscle Segmentation	1) Due to the difficulty in making a direct quantitative comparison to studies that employed full image resolution, all photos were down-sampled to 256*256 2) The actual depiction and correctness of the 832 pectoral muscle borders in a full image resolution may be impacted by image 831 down-sampling	1) The probability 835 maps produced by the region-based method could include extra information and might be utilized as a "secondary reference" to locate contours that are not evident in the probability 837 maps produced by the contour-based method
Mohammad Alkhaleefah [124]	1141 images (Not mentioned)	Not mentioned	1) Breast masses segmentation	1) One problem of histogram equalization is that, mostly because of the flattening nature of the histogram equalization, the background noise might rise when the image is too bright or too dark in the local region after the histogram equalization	1) New deep learning methods for tumor identification and classification will be implemented in future work for automated breast cancer diagnosis
Naveed Chouhan [125]	2797 images (Not mentioned)	Fivefold cross-validation	1) Detect and classify malignant and benign lesions	1) Because to inadequate information flow between network layers, deep architectures are plagued by negative learning, vanishing gradient, and feature reuse issues	1) Using both SVM and ELiEC classifiers, hybridization of all four sets of features results in greater performance

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Table 6 (continued)

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
ChuanBo Qin [126]	160 images (Not mentioned)	Not mentioned	1) Breast Tumor Segmentation	1) The results obtained by the suggested models could still not be enough for clinical use	1) This study may one day assist physicians in developing a prediction model to enhance the diagnosis, treatment, and prognosis of breast cancers by quantitatively analyzing the segmentation data from a 3D viewpoint
Li Li [127]	(96 patients)	Tenfold cross-validation	1) Prediction of Breast Cancer	1) The sample size was insufficient, and additional experimenters should have been included	1) Clinical trials have to be carried out at hospitals with several centers and big sample sizes rather than in a single or limited region
Xuehua Xiao [128]	120 images (Not mentioned)	Not mentioned	1) Detect and classify malignant and benign lesions	1) The study's small sample size 2) The multicenter, large-sample hospitals should not be the only ones used for the clinical study	1) Deep learning has been employed extensively in people's lives, including the medical sector, and its contribution to advancing the development of medical technology cannot be understated as a potential research path
Natalia Antropova [7]	1815 images (Not mentioned)	Not mentioned	1) Detect and classify malignant and benign lesions	1) Because to the depth (up to 1000 layers) and intricacy, this work is unable to investigate the possibility of CADx 2) Datasets originated from a single medical facility as well. It is unclear if our classifiers will perform well on images from another institution due to the heterogeneity brought on by various imaging manufacturers and facility methods	1) The suggested solutions to the efficiency, accuracy, and preprocessing problems open the door to more efficient CADx techniques
Ayelet Akserlod-Ballin et al. [129]	52,936 images (13,234 patient) Mean age of 56 ± 10 years	Fivefold cross-validation	1) Prediction of malignancy 2) Screening mammography	1) Women with positive biopsy results had a lower risk of breast cancer in their families	1) It is anticipated that combining clinical data as well as other aspects like genetic data in the future would improve the performance
Yeşim Eroğlu et al.[130]	780 images (Not mentioned)	Fivefold cross-validation	1) Detect and classify malignant, benign lesions and normal	1) Insufficient data in the data set	1) Using a larger dataset
Michal Byra et al. [131]	882 images (Not mentioned)	Tenfold cross-validation	1) Detect and classify malignant and benign lesions	1) The radiologist's final evaluation disregarded the CNN findings	1) It is crucial to look into the possibility of classifying benign and malignant breast subtypes using CNN

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Table 6 (continued)

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Ravi K Samala et al. [132]	2,454 masses (Not mentioned)	Fourfold cross-validation	1) Detect and classify malignant and benign lesions	1) Did not compare various layer combinations 2) It is unclear whether DCNNs like VGG and GoogleNet exhibit the same degree of adaptability in multitask transfer learning or not 3) Further data are required 4) Failing to contrast the effectiveness of DCNN with the effectiveness of other machine learning methods 5) The vendor's DM system gathered the little DM data set	1) Future research should take into account including DM from other vandors as extra tasks and comparing test results across DM sets
Songfeng Li et al. [133]	478 images (352 patients)	Tenfold cross-validation	1) Classify into fatty class or dense class	1) Data from a single vendor's mammography systems 2) Further image validation of this technique will be required To balance the general distribution of dense and fatty breasts in the training population, the percentage of fatty breasts was decreased	1) More samples, particularly dense breasts, must be used to fine-tune the DCNN
Ravi K. Samala et al. [134]	2566 views (Not mentioned)	Two-loop leave-one-case-out cross-validation	1) Detect and classify malignant and benign lesions	1) Limited data sets and the expensive computation 2) The numerous potential permutations by freezing various numbers of layers were not examined	1) Using large dataset
Dooman Arefan et al. [8]	452 images (226 patients)	Tenfold cross-validation	1) Predict short-term risk 2) Screening mammography	1) This study's sample size is small 2) A single provider provided the images	1) It would be preferable to enhance the first evaluation and utilize a wider multicenter dataset
Sarah S. Aboutalib et al. [135]	14,860 images (3715 patients)	Sixfold cross-validation	1) Detect and classify malignant and benign lesions	1) Two distinct datasets were utilized	1) Deep learning would not provide as much information as other structures like VGG, Residual Network, and GoogleLeNet
Yang Zhang et al. [136]	Not mentioned (286 patients)	Tenfold cross-validation	1) Quantify fibro glandular tissue volume in breast	1) Only the nonfat-suppressed T1-weighted MR image was examined 2) Compared to Western women, Asian women have breasts that are denser and have different physical characteristics	1) 3D segmentation may be added to the proposed U-Net architecture

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Contribution	Total number of images/masses/ examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Nan Wu et al. [137]	5447 images (Not mentioned)	Not mentioned	1) Detect and classify malignant and benign lesions	1) This architecture will disregard outside information	1) Employing the hybrid model may increase radiologists' diagnostic sensitivity
Jiejie Zhou et al. [138]	153 masses (133 patients)	Tenfold cross-validation	1) Detect and classify malignant and benign lesions	1) Limited data set 2) To prevent overfitting, appropriate techniques like L2 norm regularization and dropout were applied.	1) Utilized with appropriate transfer learning on other datasets
Arun Asokan Nair et al. [139]	22 230 images	Not mentioned	1) Breast Tumor Segmentation	1) Smaller cysts were more difficult to treat than bigger ones	1) Researching the advantages of generating many outputs from a single input for decision-making is preferable
Simukayi Mutasa et al. [140]	246 images (123 patients)	Fivefold cross-validation	1) Predict pure Ductal Carcinoma in Situ (DCIS) versus DCIS with invasion	1) The retrospective research conducted at a single institution and the small sample size 2) The interpretation process was not carried out by many pathologists	Not mentioned
Richard Ha et al. [141]	1474 images (210 patients)	Fivefold cross-validation	1) Predicts risk for breast cancer	1) It is retrospective research at a single institution with a very limited dataset 2) Long training times are an inherent restriction of CNN, and the classification of breast density might be arbitrary and sensitive to intra- and inter-observer heterogeneity	1) A prospective randomized trial is required for validation
Michal Byra et al. [142]	893 images (Not mentioned)	Fivefold cross-validation	1) Breast Tumor Segmentation	1) Did not evaluate the utility of various deep learning-based segmentation networks 2) Four datasets' ROIs were created by various medical experts 3) Manual segmentation's accuracy is not entirely known	1) Transfer learning and incorporating additional data might enhance segmentation performance
Yang Zhang et al. [143]	(244 patients)	Tenfold cross-validation	1) Differentiate three breast cancer molecular subtypes	1) The small case number was a severe consequence that often led to poor accuracy, especially for the TN subtype	1) Based on the quantity of patients, clinical signs, and relevant datasets, applied ML for assessing medical images may be built

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Table 6 (continued)

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Siyuan Lu et al. [144]	813 images	Fivefold cross-validation	1) Detect and classify malignant and benign lesions	1) The existing dataset is limited	1) In order to use vision transformers for breast cancer diagnosis employing optimization techniques to train the RNNs and segmentation, more breast USIs and the proper dataset must be collected
Viksit Kumar et al. [145]	258 images (433 patients)	Tenfold cross-validation	1) Breast Tumor Segmentation	1) The in vivo tissue can only be seen planarly in clinical 2D US images 1) No combining of data from various cross sections is done 1) The tiny training set makes it difficult to separate unique diseases, which may combine two distinct pathologies	1) Using a bigger dataset
Chen Zhang [146]	10,480 images (2620 patients)	Tenfold cross-validation	1) Classification (Normal/Abnormal and Benign/Malignancy)	1) A limited sample size, 2) The feature fusion technique is not widely discussed	1) The model's performance must be enhanced by transfer based on comparable data, and a more successful feature fusion approach must be explored
Maleika Heenaye-Mamode Khan et al. [147]	2620 images (Not mentioned)	k-fold cross-validation	1) Detect and classify malignant and benign lesions	1) Limited dataset	1) The methods for diagnosing and categorizing breast abnormalities must be determined, as well as the best imaging methods and treatments for this group
Tariq Mahmood et al. [24]	3341 images (Not mentioned)	Fivefold cross-validation	1) Detect and classify benign and malignant tumors	1) Failed to comprehend the degree of difficulties brought on by different types of breast cancer 2) There is a severe paucity of data from mammography images	1) Future research must make use of more comprehensive data, mammography images, and collections
Feng-Zhou Zhang et al. [148]	156 images (87 patients)	Tenfold cross-validation	1) Breast Lesion Screening	Not mentioned	1) By adding new clinical imaging datasets continually, classification performance may be enhanced in the future to distinguish tumors that are B or M

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Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Meng Zhu et al. [149]	568 images (2492 patients)	Not mentioned	1) Identify ductal carcinoma in situ and microinvasion	1) Mammography data were not utilized; only ultrasound data were 2) Data from MRI were not included 3. The dataset was little 4) Since it was retrospective research, there were an unequal amount of ultrasound images 5) Due to the lack of clinicopathological data, a logistic regression model was not used for the external validation	1) For more thorough features, it is preferable to employ fully automated volume imaging and three-dimensional ultrasound
Daniel Truhn et al. [150]	1294 images (447 patients)	Tenfold cross-validation	1) Classification of contrast-enhancing lesions at multi-parametric breast	1) In the research, sensitivity refers to correctly classifying lesions 2) MR images were collected using a consistent methodology, while pulse sequence methods vary across institutions 3) The accuracy of the trained algorithms may not be as high when applied to data from different acquisition techniques 4) This algorithm only used a subset of three	1) Applying bigger dataset
Adam Yala et al. [151]	88,994 images (39 571 patients)	Not mentioned	1) Improved Breast Cancer Risk Prediction	1) It employed mammograms taken by a single vendor and patient data from a single tertiary academic institution (Hologic). Information on risk factors was unavailable for certain patients	1) More research must be done to verify the model
Adam Yala et al. [152]	238 271 images (66 661 patients)	Not mentioned	1) Triage Screening Mammograms	1) This investigation of the possible effects of applying the model to mammography triage was done in the past 2) Only a drop in sensitivity or an increase in specificity may be shown using this simulation method	1) Conducting a prospective trial is necessary to ascertain the model's effect on radiologists, as well as to assess the model's effectiveness across various suppliers and to conduct the necessary external validation with distinct populations

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Table 6 continued

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Li-Qiang Zhou et al. [153]	964 images (756 patients)	Not mentioned	1) Lymph node metastasis prediction	1) The research was retrospective and included a little amount of data 2) There was some variation in the visual quality 3) The diagnosis of lymph node metastases and the absence of metastasis depended on the timing of breast surgery	1) Larger, prospective studies were conducted before real clinical usage, which is preferable
Ridhi Arora et al. [154]	2620 images (1318 patients)	Not mentioned	1) Detect and classify malignant and benign lesions	Not mentioned	1) Segmenting ROIs from larger datasets is essential
Revathy Sivanandan, Jayakumari. [155]	580 images (Not mentioned)	Fivefold cross-validation	1) Detect and classify malignant and benign lesions	1) Clustered activation maps failed to capture severe isoechoic lesions	1) By incorporating comparable sample layers into CNN, machine-based tumor semantic segmentation must be carried out
Mohammed El Adoui et al. [156]	763 images (43 patients)	Fourfold cross-validation	1) Predicting breast tumor response to chemotherapy	1) The dataset is little in size 2) This method requires a series of manual data preprocessing steps	1) This ML model must be included into an interactive web application and used with further datasets
Tomoyuki Fujioka et al. [157]	947 images (237 patients)	Not mentioned	1) Detect and classify malignant and benign lesions	1) A single institution conducted retrospective research 2) Not all recurring lesions were identified by cytological or histological examination 3) Images may lose some information while processing 4) There could have been an issue with the three ultrasound systems' capacity to adjust	1) It is preferable to confirm that ultrasonic systems can work well
Haibo Huang et al. [158]	563 images (108 patients)	Not mentioned	1) Breast masses segmentation	Not mentioned	1) Further research on the clinical applicability of this model is required
Huiling Xiang et al. [159]	444 images (396 patients)	Fivefold cross-validation	1) Detect and classify malignant and benign lesions	1) The demand for computing resources 2) Small dataset size	1) To get over the restrictions and enhance system performance, various CNN architectures and ML should be applied
Lu Huo et al. [160]	(100 patients)	Fivefold cross-validation	1) Breast masses segmentation and fibro glandular tissue	1) The FGT segmentation may be impacted by the outcomes of the breast segmentation 2) The 3D segmentation inferencing time in this study was around 40 s	1) A bigger, multicenter data set must be utilized for the model to be reliable
Thomas Ren et al. [161]	238 images (Not mentioned)	Threefold cross-validation	1) Detect nodal metastasis with 18FDG-PET as ground truths	1) Sample selection bias that was not deliberate may exist 2) The utilization of ground truth from 18 FDG PET 3) Avoiding the biopsied nodes' artifacts could lead to bias	1) To achieve generalizability, prospective, multicenter research in which MRI is collected before aLN dissection of all pertinent nodes for pathology is required

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Table 6 (continued)

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Jan Witowski et al. [162]	21,537 images (13,463 patients)	Not mentioned	1) Detect and classify malignant and benign lesions	1) The DL system could influence the radiologist's decision-making 2) A cost-effective analysis for the DL system should be conducted ignore the fact that the system has learnt to exploit small hints in images to influence its decisions	1) To determine the financial advantages and reasonable expenses of the DL system, a more thorough cost-effective analysis should be conducted
Fei Gao et al. [163]	118 images. [49 CEDM cases69 DM cases]	Twofold cross-validation	1) Detect and classify malignant and benign lesions	Not mentioned	1) It is preferable to determine if the deeper layers demonstrate morphological features
Mohammad I. Daoud et al. [164]	380 images (Not mentioned)	Tenfold cross-validation	1) Detect and classify malignant and benign lesions	Not mentioned	1) In order to increase the classification performance, it is preferable to combine the deep characteristics of the BUS images using several CNN models
Khalil ur Rehman et al. [165]	577 images (289 patients)	Fivefold cross-validation	1) Detect and classify malignant and benign lesions	Not mentioned	1) To enhance classification and data imbalance, it is preferable to explore the identification of individual MCs, use alternative DL to classify on a big dataset, and assess Sensors models
Kuen-Jang Tsai et al. [166]	5733 images (1490 patients)	Not mentioned	1) Detect and classify malignant and benign lesions	Not mentioned	1) Using bigger dataset
Wei-Chung Shia et al. [167]	684 images (343 patients)	Fivefold cross-validation	1) Breast masses segmentation 2) Prediction malignancy in breast	One) Limited sample size 2) The issue with replicating the FCN-32s' semantic segmentation performance 3) The segmented region's output may sometimes be inconsistent with the actual data	1) It seems that the creation of advanced designs or the modification of network topologies may address these restrictions
Haolin Yin et al. [168]	136 images (136 patients)	Tenfold cross-validation	1) Preoperative assessment of breast cancer molecular subtypes	1) This retrospective research may have selection bias, 2) Not validated by formal genetic testing 3) The proportions of the four molecular subtypes in IHC surrogates were out of balance	1) A bigger dataset must be used in order to extend the model and choose an appropriate sequence
Kuochen Zhou et al. [169]	322 images	Not mentioned	1) To extract breast region accurately from mammographic images	1) Insufficient number of images for evaluation	1) This model has to be tested with a bigger data set
Yi Wang et al. [170]	614 images (219 patients)	Fourfold cross-validation	1) Detect and classify malignant and benign lesions	1) This approach could overlook certain cancer cases and wrongly treat the cancerous spots as ordinary ultrasonography shadows	1) A technique for automatically identifying benign and malignant tumors must be created

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Table 6 (continued)

Contribution	Total number of images/masses/examinations/views (number of patients/cases)	Validation method	Application of technique	Limitation and obstacles of implementation	Future directions
Xin Shu et al. [171]	3481 images (Not mentioned)	Not mentioned	1) Detect and classify malignant and benign lesions	1) Weak supervisory learning, which requires no matching labels at all, is the method through which the capacity to locate lesions is gained	1) To achieve a better model on various datasets and enhance this model, the k value must be manually set
Manal AlGhamdi et al. [172]	1520 images (1,700 patients)	Fivefold cross-validation	1) Detection of Arterial Calcification	1) A little, poor quality dataset	1) Additional methods for detecting BACs need to be developed, tested, and used, along with a bigger dataset for training with a wide range of cases
Masoomeh Rahimpour et al. [173]	141 images (Not mentioned)	Not mentioned	1) Breast masses segmentation	1) The datasets available in the database were few 2) The suggested technique is not totally automated, necessitating an extra visual evaluation	1) More post-contrast image uses should be researched, along with the addition of other modalities including diffusion-weighted images and apparent diffusion coefficient maps
Juan Zhou et al. [174]	(1031 patients)	Not mentioned	1) Detect and classify malignant and benign lesions	1) Only MRIs with a single lesion and instances where BPE is obvious are included 2) Just structural characteristics were taken into account while training the network; no particular anomalies were trained for	1) Image-level monitoring alone has not yet been extensively researched for the direct detection of breast cancer in 3D images
Yuchen Qiu et al. [175]	560 ROIs	Fourfold cross-validation	1) Detect and classify malignant and benign lesions	1) Only DL was assessed 2) The deep learning-based CAD method is still in its infancy 3) The size of the dataset is limited	1) It seems that further deep learning networks have been researched that employ 293 new or more sophisticated layers

Acknowledgements Not applicable.

Author contributions Authors SR/MZ/LSH/MN wrote the first draft of the manuscript. Authors SR/EE/NR/GA performed data collection and analysis and extracted main characteristics. All authors SR/MN/EE/NR/GA/LSH/MZ/HA reviewed and provided critical feedback. All authors read and approved the final manuscript.

Funding In this paper, we did not have any financial sponsors.

Data availability All data generated or analyzed during this study is included in this published article and its supplementary information files.

Declarations

Conflict of interest The authors declare that there is no conflict of interest regarding the publication of this article.

Ethical approval All methods were carried out in accordance with relevant guidelines and regulations. The methodology for this study was approved by the ethics committee of Tehran University of Medical Sciences (Ethical code IR.TUMS.S-PH.REC.1401.137, Research code: 1401-2-102-59273).

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Human and animal rights Since there were no human or animal subjects used in this study, there is no need for ethical approval.

Informed consent As there were no subjects in this study, there was no need for consent because there was no informal participation.

References

1. Fahad UM (2019) Breast cancer: current perspectives on the disease status. Adv Exp Med Biol 1152:51–64
2. Ragb H, Ali R, Jera E, Buaossa N (2021) Convolutional neural network based on transfer learning for breast cancer screening. arXiv preprint arXiv:211211629
3. Joshi RC, Singh D, Tiwari V, Dutta MK (2022) An efficient deep neural network based abnormality detection and multi-class breast tumor classification. Multimed Tools Appl 81(10):13691–13711
4. Siegel RL, Miller KD, Fuchs HE, Jemal A (2022) Cancer statistics, 2022. CA Cancer J Clin 72(1):7–33
5. Waks AG, Winer EP (2019) Breast cancer treatment: a review. JAMA 321(3):288–300
6. Ahmed L, Iqbal MM, Aldabbas H, Khalid S, Saleem Y, Saeed S (2020) Images data practices for semantic segmentation of breast cancer using deep neural network. J Amb Intell Humaniz Comput 14:1–17
7. Antropova N, Huynh BQ, Giger ML (2017) A deep feature fusion methodology for breast cancer diagnosis demonstrated on three imaging modality datasets. Med Phys 44(10):5162–5171
8. Arefan D, Mohamed AA, Berg WA, Zuley ML, Sumkin JH, Wu S (2020) Deep learning modeling using normal mammograms for predicting breast cancer risk. Med Phys 47(1):110–118
9. Agaian S, Madhukar M, Chronopoulos AT (2014) Automated screening system for acute myelogenous leukemia detection in blood microscopic images. IEEE Syst J 8(3):995–1004
10. Luo L, Wang X, Lin Y, Ma X, Tan A, Chan R, et al. (2024) Deep learning in breast cancer imaging: A decade of progress and future directions. IEEE Rev Biomed Eng.
11. Mohapatra S, Patra D, editors (2010) Automated cell nucleus segmentation and acute leukemia detection in blood microscopic images. In: 2010 International Conference on Systems in Medicine and Biology. IEEE.
12. Selvathi D, Aarthy PA (2018) Deep learning techniques for breast cancer detection using medical image analysis. Springer, Biologically rationalized computing techniques for image processing applications, pp 159–186
13. Yassin NI, Omran S, El Houby EM, Allam H (2018) Machine learning techniques for breast cancer computer aided diagnosis using different image modalities: a systematic review. Comput Methods Programs Biomed 156:25–45
14. Arevalo J, Cruz-Roa A, González FA,Eds. (2013) Hybrid image representation learning model with invariant features for basal cell carcinoma detection. In: IX international seminar on medical information processing and analysis. SPIE
15. Chi C-L, Street WN, Wolberg WH, Eds. (2007). Application of artificial neural network-based survival analysis on two breast cancer datasets. In: AMIA annual symposium proceedings. American Medical Informatics Association
16. Jadoon MM, Zhang Q, Haq IU, Butt S, Jadoon A (2017) Three-class mammogram classification based on descriptive CNN features. BioMed Res Int 2017:3640901
17. Campana D, Coustan-Smith E (1999) Detection of minimal residual disease in acute leukemia by flow cytometry. Cytomet: J Int Soc Anal Cytol. 38(4):139–52
18. Sahu A, Das PK, Meher S (2024) An efficient deep learning scheme to detect breast cancer using mammogram and ultrasound breast images. Biomed Signal Process Control 87:105377
19. Aggarwal R, Sounderajah V, Martin G, Ting DS, Karthikesalingam A, King D et al (2021) Diagnostic accuracy of deep learning in medical imaging: a systematic review and meta-analysis. NPJ digital medicine 4(1):65
20. Liu H, Xie X, Wang B (2024) Deep learning infers clinically relevant protein levels and drug response in breast cancer from unannotated pathology images. NPJ Breast Cancer 10(1):18
21. Giger ML (2018) Machine learning in medical imaging. J Am Coll Radiol 15(3):512–520
22. Nilashi M, Ahmadi H, Manaf AA, Rashid TA, Samad S, Shahmoradi L et al (2020) Coronary heart disease diagnosis through self-organizing map and fuzzy support vector machine with incremental updates. Int J Fuzzy Syst 22(4):1376–1388
23. Nilashi M, Ahmadi H, Shahmoradi L, Mardani A, Ibrahim O, Yadegaridehkordi E (2017) Knowledge discovery and diseases prediction: a comparative study of machine learning techniques. J Soft Comput & Dec Support Syst 4(5):8
24. Mahmood T, Li J, Pei Y, Akhtar F, Rehman MU, Wasti SH (2022) Breast lesions classifications of mammographic images using a deep convolutional neural network-based approach. PLoS ONE 17(1):e0263126
25. Shahmoradi L, Safdari R, Ahmadi H, Zahmatkeshan M (2021) Clinical decision support systems-based interventions to improve medication outcomes: a systematic literature review on features and effects. Med J Islam Repub Iran 35:27
26. Rajkomar A, Dean J, Kohane I (2019) Machine learning in medicine. N Engl J Med 380(14):1347–1358

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27. You J, Huang Y, Ouyang L, Zhang X, Chen P, Wu X et al (2024) Automated and reusable deep learning (AutoRDL) framework for predicting response to neoadjuvant chemotherapy and axillary lymph node metastasis in breast cancer using ultrasound images: a retrospective, multicentre study. eClinicalMedicine 69:102499
28. Omonigho EL, David M, Adejo A, Aliyu S, Eds (2020) Breast cancer: tumor detection in mammogram images using modified alexnet deep convolution neural network. In: 2020 International Conference in Mathematics, Computer Engineering and Computer Science (ICMCECS). IEEE.
29. Shahmoradi L, Farzanehnejad AR, Goli A (2016) Guideline-based clinical decision support systems as an inseparable tool for better cancer care management. Iran J Public Health 45(7):962
30. Houssein EH, Emam MM, Ali AA, Suganthan PN (2021) Deep and machine learning techniques for medical imaging-based breast cancer: a comprehensive review. Expert Syst Appl 167:114161
31. Niyaz U, Sambyal AS, Eds (2018) Advances in deep learning techniques for medical image analysis. In: 2018 Fifth International Conference on Parallel, Distributed and Grid Computing (PDGC). IEEE
32. Chorianopoulos AM, Daramouskas I, Perikos I, Grivokostopoulou F, Hatzilygeroudis I, Eds (2020) Deep learning methods in medical imaging for the recognition of breast cancer. In: 2020 11th International Conference on Information, Intelligence, Systems and Applications (IISA). IEEE.
33. Zou L, Yu S, Meng T, Zhang Z, Liang X, Xie Y (2019) A technical review of convolutional neural network-based mammographic breast cancer diagnosis. Comput Math Methods Med 2019:6509357
34. Fujioka T, Mori M, Kubota K, Oyama J, Yamaga E, Yashima Y et al (2020) The utility of deep learning in breast ultrasonic imaging: a review. Diagnostics 10(12):1055
35. Jiménez-Gaona Y, Rodríguez-Álvarez MJ, Lakshminarayanan V (2020) Deep-learning-based computer-aided systems for breast cancer imaging: a critical review. Appl Sci 10(22):8298
36. Madani M, Behzadi MM, Nabavi S (2022) The role of deep learning in advancing breast cancer detection using different imaging modalities: a systematic review. Cancers 14(21):5334
37. Nasser M, Yusof UK (2023) Deep learning based methods for breast cancer diagnosis: a systematic review and future direction. Diagnostics 13(1):161
38. Mridha MF, Hamid M, Monowar MM, Keya AJ, Ohi AQ, Islam M et al (2021) A comprehensive survey on deep-learning-based breast cancer diagnosis. Cancers 13(23):6116
39. Xue P, Wang J, Qin D, Yan H, Qu Y, Seery S et al (2022) Deep learning in image-based breast and cervical cancer detection: a systematic review and meta-analysis. NPJ Digit Med 5(1):19
40. Murtaza G, Shuib L, Abdul Wahab AW, Mujtaba G, Mujtaba G, Nweke HF et al (2020) Deep learning-based breast cancer classification through medical imaging modalities: state of the art and research challenges. Artif Intell Rev 53:1655–1720
41. Fatima N, Liu L, Hong S, Ahmed H (2020) Prediction of breast cancer, comparative review of machine learning techniques, and their analysis. IEEE Access 8:150360–150376
42. Westphaln KK, Regoeczi W, Masotya M, Vazquez-Westphaln B, Lounsbury K, McDavid L et al (2021) From Arksey and O'Malley and Beyond: customizations to enhance a team-based, mixed approach to scoping review methodology. MethodsX 8:101375
43. Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD et al (2021) The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. BMJ (Clin Res ed) 372:n71
44. Duggento A, Aiello M, Cavaliere C, Cascella GL, Cascella D, Conte G et al (2019) An ad hoc random initialization deep neural network architecture for discriminating malignant breast cancer lesions in mammographic images. Contrast Media & Mol Imaging 2019:5982834
45. Cai H, Huang Q, Rong W, Song Y, Li J, Wang J et al (2019) Breast microcalcification diagnosis using deep convolutional neural network from digital mammograms. Comput Math Methods Med 2019:2717454
46. Jiao H, Jiang X, Pang Z, Lin X, Huang Y, Li L (2020) Deep convolutional neural networks-based automatic breast segmentation and mass detection in DCE-MRI. Comput Math Methods Med 2020:2431706
47. Lai X, Yang W, Li R (2020) DBT masses automatic segmentation using U-Net neural networks. Comput Math Methods Med 2020:7156165
48. Ramadan SZ (2020) Using convolutional neural network with cheat sheet and data augmentation to detect breast cancer in mammograms. Comput Math Methods Med. 2020:9523404
49. El Adoui M, Mahmoudi SA, Larhmam MA, Benjelloun M (2019) MRI breast tumor segmentation using different encoder and decoder CNN architectures. Computers 8(3):52
50. Albalawi U, Manimurugan S, Varatharajan R (2022) Classification of breast cancer mammogram images using convolution neural network. Concurr Comput: Pract Exp 34(13):e5803
51. Rouhi R, Jafari M (2016) Classification of benign and malignant breast tumors based on hybrid level set segmentation. Expert Syst Appl 46:45–59
52. Caballo M, Pangallo DR, Mann RM, Sechopoulos I (2020) Deep learning-based segmentation of breast masses in dedicated breast CT imaging: radiomic feature stability between radiologists and artificial intelligence. Comput Biol Med 118:103629

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53. Ribli D, Horváth A, Unger Z, Pollner P, Csabai I (2018) Detecting and classifying lesions in mammograms with deep learning. Sci Rep 8(1):4165
54. Hassanien MA, Singh VK, Puig D, Abdel-Nasser M (2022) Predicting breast tumor malignancy using deep ConvNeXt radiomics and quality-based score pooling in ultrasound sequences. Diagnostics 12(5):1053
55. Sabani A, Landsmann A, Hejduk P, Schmidt C, Marcon M, Borkowski K et al (2022) BI-RADS-based classification of mammographic soft tissue opacities using a deep convolutional neural network. Diagnostics 12(7):1564
56. Shenbagavalli P, Thangarajan R (2018) Aiding the digital mammogram for detecting the breast cancer using shearlet transform and neural network. Asian Pac J Cancer Prevent: APJCP 19(9):2665
57. Xue H, Qian G, Wu X, Gao Y, Yang H, Liu M et al (2022) A coarse-to-fine and automatic algorithm for breast diagnosis on multi-series MRI images. Front Comput Sci 4:1054158
58. Bai J, Jin A, Wang T, Yang C, Nabavi S (2022) Feature fusion Siamese network for breast cancer detection comparing current and prior mammograms. Med Phys 49(6):3654–3669
59. Yin H, Jiang Y, Xu Z, Huang W, Chen T, Lin G (2022) Apparent diffusion coefficient-based convolutional neural network model can be better than sole diffusion-weighted magnetic resonance imaging to improve the differentiation of invasive breast cancer from breast ductal carcinoma in situ. Front Oncol 11:5838
60. Xu Z, Yang Q, Li M, Gu J, Du C, Chen Y et al (2022) Predicting HER2 status in breast cancer on ultrasound images using deep learning method. Front Oncol 12:829041
61. Li Y, Gu H, Wang H, Qin P, Wang J (2022) BUSnet: A deep learning model of breast tumor lesion detection for ultrasound images. Front Oncol 12:848271
62. Chen Y, Wang L, Luo R, Wang S, Wang H, Gao F et al (2022) A deep learning model based on dynamic contrast-enhanced magnetic resonance imaging enables accurate prediction of benign and malignant breast lessons. Front Oncol 12:943415
63. Zhu J, Geng J, Shan W, Zhang B, Shen H, Dong X et al (2022) Development and validation of a deep learning model for breast lesion segmentation and characterization in multiparametric MRI. Front Oncol 12:946540
64. Zhou Y, Feng B-J, Yue W-W, Liu Y, Xu Z-F, Xing W, et al. (2022) Differentiating non-lactating mastitis and malignant breast tumors by deep-learning based AI automatic classification system: A preliminary study. Front Oncol 5023
65. Li C, Guo Y, Jia L, Yao M, Shao S, Chen J, et al. (2022) A convolutional neural network based on ultrasound images of primary breast masses: prediction of lymph-node metastasis in collaboration with classification of benign and malignant tumors. Front Physiol 1058
66. Pawar SD, Sharma KK, Sapate SG, Yadav GY, Alroobaea R, Alzahrani SM et al (2022) Multichannel DenseNet architecture for classification of mammographic breast density for breast cancer detection. Front Public Health 10:885212
67. Al-Antari MA, Al-Masni MA, Choi M-T, Han S-M, Kim T-S (2018) A fully integrated computer-aided diagnosis system for digital X-ray mammograms via deep learning detection, segmentation, and classification. Int J Med Informatics 117:44–54
68. Piantadosi G, Sansone M, Fusco R, Sansone C (2020) Multi-planar 3D breast segmentation in MRI via deep convolutional neural networks. Artif Intell Med 103:101781
69. Li H, Zhuang S, Li D-a, Zhao J, Ma Y (2019) Benign and malignant classification of mammogram images based on deep learning. Biomed Signal Process Control 51:347–54
70. Soulami KB, Kaabouch N, Saidi MN, Tamtaoui A (2021) Breast cancer: One-stage automated detection, segmentation, and classification of digital mammograms using UNet model based-semantic segmentation. Biomed Signal Process Control 66:102481
71. Ren T, Cattell R, Duanmu H, Huang P, Li H, Vanguri R et al (2020) Convolutional neural network detection of axillary lymph node metastasis using standard clinical breast MRI. Clin Breast Cancer 20(3):e301–e308
72. Moon WK, Huang Y-S, Hsu C-H, Chien T-YC, Chang JM, Lee SH et al (2020) Computer-aided tumor detection in automated breast ultrasound using a 3-D convolutional neural network. Comput Methods Programs Biomed 190:105360
73. Al-Antari MA, Han S-M, Kim T-S (2020) Evaluation of deep learning detection and classification towards computer-aided diagnosis of breast lesions in digital X-ray mammograms. Comput Methods Programs Biomed 196:105584
74. AlGhamdi M, Abdel-Mottaleb M (2021) DV-DCNN: Dual-view deep convolutional neural network for matching detected masses in mammograms. Comput Methods Programs Biomed 207:106152
75. Qi X, Zhang L, Chen Y, Pi Y, Chen Y, Lv Q et al (2019) Automated diagnosis of breast ultrasonography images using deep neural networks. Med Image Anal 52:185–198
76. Shi J, Zhou S, Liu X, Zhang Q, Lu M, Wang T (2016) Stacked deep polynomial network based representation learning for tumor classification with small ultrasound image dataset. Neurocomputing 194:87–94
77. Pan P, Chen H, Li Y, Cai N, Cheng L, Wang S (2021) Tumor segmentation in automated whole breast ultrasound using bidirectional LSTM neural network and attention mechanism. Ultrasonics 110:106271
78. Fan M, Li Y, Zheng S, Peng W, Tang W, Li L (2019) Computer-aided detection of mass in digital breast tomosynthesis using a faster region-based convolutional neural network. Methods 166:103–111

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79. Byra M, Dobruch-Sobczak K, Klimonda Z, Piotrzkowska-Wroblewska H, Litniewski J (2020) Early prediction of response to neoadjuvant chemotherapy in breast cancer sonography using Siamese convolutional neural networks. IEEE J Biomed Health Inform 25(3):797–805
80. Singla N, Dubey K, Srivastava V (2019) Automated assessment of breast cancer margin in optical coherence tomography images via pretrained convolutional neural network. J Biophotonics 12(3):e201800255
81. Antropova N, Huynh B, Li H, Giger ML (2019) Breast lesion classification based on dynamic contrast-enhanced magnetic resonance images sequences with long short-term memory networks. J Med Imaging 6(1):011002
82. Wang Y, Wang S, Chen J, Wu C (2020) Whole mammographic mass segmentation using attention mechanism and multiscale pooling adversarial network. J Med Imaging 7(5):054503
83. Wang F, Liu X, Yuan N, Qian B, Ruan L, Yin C et al (2020) Study on automatic detection and classification of breast nodule using deep convolutional neural network system. J Thorac Dis 12(9):4690
84. Zhou B-Y, Wang L-F, Yin H-H, Wu T-F, Ren T-T, Peng C et al (2021) Decoding the molecular subtypes of breast cancer seen on multimodal ultrasound images using an assembled convolutional neural network model: a prospective and multicentre study. EBioMedicine 74:103684
85. Chougrad H, Zouaki H, Alheyane O (2018) Deep convolutional neural networks for breast cancer screening. Comput Methods Programs Biomed 157:19–30
86. Samala RK, Chan HP, Hadjiiski L, Helvie MA, Wei J, Cha K (2016) Mass detection in digital breast tomosynthesis: deep convolutional neural network with transfer learning from mammography. Med Phys 43(12):6654–6666
87. Chiang T-C, Huang Y-S, Chen R-T, Huang C-S, Chang R-F (2018) Tumor detection in automated breast ultrasound using 3-D CNN and prioritized candidate aggregation. IEEE Trans Med Imaging 38(1):240–249
88. Wang Y, Choi EJ, Choi Y, Zhang H, Jin GY, Ko S-B (2020) Breast cancer classification in automated breast ultrasound using multiview convolutional neural network with transfer learning. Ultrasound Med Biol 46(5):1119–1132
89. Diniz JOB, Diniz PHB, Valente TLA, Silva AC, de Paiva AC, Gattass M (2018) Detection of mass regions in mammograms by bilateral analysis adapted to breast density using similarity indexes and convolutional neural networks. Comput Methods Programs Biomed 156:191–207
90. Evain E, Raynaud C, Ciofolo-Veit C, Popoff A, Caramella T, Kbaier P et al (2021) Breast nodule classification with two-dimensional ultrasound using Mask-RCNN ensemble aggregation. Diagn Interv Imaging 102(11):653–658
91. Yang H, Yang D (2023) CSwin-PNet: A CNN-Swin transformer combined pyramid network for breast lesion segmentation in ultrasound images. Expert Syst Appl 213:119024
92. Cho SW, Baek NR, Park KR (2022) Deep learning-based multi-stage segmentation method using ultrasound images for breast cancer diagnosis. J King Saud Univ-Comput Inf Sci 34(10):10273–10292
93. Byra M (2021) Breast mass classification with transfer learning based on scaling of deep representations. Biomed Signal Process Control 69:102828
94. Jiang M, Zhang D, Tang S-C, Luo X-M, Chuan Z-R, Lv W-Z et al (2021) Deep learning with convolutional neural network in the assessment of breast cancer molecular subtypes based on US images: a multicenter retrospective study. Eur Radiol 31:3673–3682
95. Altaf MM (2021) A hybrid deep learning model for breast cancer diagnosis based on transfer learning and pulse-coupled neural networks. Math Biosci Eng 18(5):5029–5046
96. Singh VK, Rashwan HA, Romani S, Akram F, Pandey N, Sarker MMK et al (2020) Breast tumor segmentation and shape classification in mammograms using generative adversarial and convolutional neural network. Expert Syst Appl 139:112855
97. Cao X, Chen H, Li Y, Peng Y, Zhou Y, Cheng L et al (2022) Auto-DenseUNet: Searchable neural network architecture for mass segmentation in 3D automated breast ultrasound. Med Image Anal 82:102589
98. Al-Masni MA, Al-Antari MA, Park J-M, Gi G, Kim T-Y, Rivera P et al (2018) Simultaneous detection and classification of breast masses in digital mammograms via a deep learning YOLO-based CAD system. Comput Methods Programs Biomed 157:85–94
99. Niu J, Li H, Zhang C, Li D (2021) Multi-scale attention-based convolutional neural network for classification of breast masses in mammograms. Med Phys 48(7):3878–3892
100. Su Y, Liu Q, Xie W, Hu P (2022) YOLO-LOGO: A transformer-based YOLO segmentation model for breast mass detection and segmentation in digital mammograms. Comput Methods Programs Biomed 221:106903
101. Ha R, Chang P, Mema E, Mutasa S, Karcich J, Wynn RT et al (2019) Fully automated convolutional neural network method for quantification of breast MRI fibroglandular tissue and background parenchymal enhancement. J Digit Imaging 32:141–147
102. Ha R, Mutasa S, Karcich J, Gupta N, Pascualvansant E, Nemer J et al (2019) Predicting breast cancer molecular subtype with MRI dataset utilizing convolutional neural network algorithm. J Digit Imaging 32:276–82
103. Zhang H, Han L, Chen K, Peng Y, Lin J (2020) Diagnostic efficiency of the breast ultrasound computer-aided prediction model based on convolutional neural network in breast cancer. J Digit Imaging 33:1218–1223
104. Hizukuri A, Nakayama R, Nara M, Suzuki M, Namba K (2021) Computer-aided diagnosis scheme for distinguishing between benign and malignant masses on breast DCE-MRI images using deep convolutional neural network with Bayesian optimization. J Digit Imaging 34:116–123

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105. Huang Y, Han L, Dou H, Luo H, Yuan Z, Liu Q et al (2019) Two-stage CNNs for computerized BI-RADS categorization in breast ultrasound images. Biomed Eng Online 18(1):1–18
106. Ma H, Tian R, Li H, Sun H, Lu G, Liu R et al (2021) Fus2Net: a novel convolutional neural network for classification of benign and malignant breast tumor in ultrasound images. Biomed Eng Online 20(1):1–15
107. Gu Y, Xu W, Lin B, An X, Tian J, Ran H et al (2022) Deep learning based on ultrasound images assists breast lesion diagnosis in China: a multicenter diagnostic study. Insights Imaging 13(1):124
108. Kim J, Kim HJ, Kim C, Lee JH, Kim KW, Park YM et al (2021) Weakly-supervised deep learning for ultrasound diagnosis of breast cancer. Sci Rep 11(1):24382
109. Taleghamar H, Jalalifar SA, Czarnota GJ, Sadeghi-Naini A (2022) Deep learning of quantitative ultrasound multiparametric images at pre-treatment to predict breast cancer response to chemotherapy. Sci Rep 12(1):1–13
110. Praveen SP, Srinivasu PN, Shafi J, Wozniak M, Ijaz MF (2022) ResNet-32 and FastAI for diagnoses of ductal carcinoma from 2D tissue slides. Sci Rep 12(1):20804
111. Zhang X, Zhang Y, Han EY, Jacobs N, Han Q, Wang X et al (2018) Classification of whole mammogram and tomosynthesis images using deep convolutional neural networks. IEEE Trans Nanobiosci 17(3):237–242
112. Fujioka T, Katsuta L, Kubota K, Mori M, Kikuchi Y, Kato A et al (2020) Classification of breast masses on ultrasound shear wave elastography using convolutional neural networks. Ultrason Imaging 42(4–5):213–220
113. Wijata A, Andrzejewski J, Pyciński B (2021) An automatic biopsy needle detection and segmentation on ultrasound images using a convolutional neural network. Ultrason Imaging 43(5):262–272
114. Frazer HM, Qin AK, Pan H, Brotchie P (2021) Evaluation of deep learning-based artificial intelligence techniques for breast cancer detection on mammograms: Results from a retrospective study using a BreastScreen Victoria dataset. J Med Imaging Radiat Oncol 65(5):529–537
115. Yap MH, Pons G, Marti J, Ganau S, Sentis M, Zwiggelaar R et al (2017) Automated breast ultrasound lesions detection using convolutional neural networks. IEEE J Biomed Health Inform 22(4):1218–1226
116. Arevalo J, González FA, Ramos-Pollán R, Oliveira JL, Lopez MAG (2016) Representation learning for mammography mass lesion classification with convolutional neural networks. Comput Methods Programs Biomed 127:248–257
117. Ragab M, Albukhari A, Alyami J, Mansour RF (2022) Ensemble deep-learning-enabled clinical decision support system for breast cancer diagnosis and classification on ultrasound images. Biology 11(3):439
118. Ciritsis A, Rossi C, Vittoria De Martini I, Eberhard M, Marcon M, Becker AS et al (2019) Determination of mammographic breast density using a deep convolutional neural network. British J Radiol 92(1093):20180691
119. Xiao T, Liu L, Li K, Qin W, Yu S, Li Z (2018) Comparison of transferred deep neural networks in ultrasonic breast masses discrimination. BioMed Res Int 2018:4605191
120. Ashokkumar N, Meera S, Anandan P, Murthy MYB, Kalaivani K, Alahmadi TA et al (2022) Deep learning mechanism for predicting the axillary lymph node metastasis in patients with primary breast cancer. BioMed Res Int 2022:8616535
121. Zeimarani B, Costa MGF, Nurani NZ, Bianco SR, Pereira WCDA, Costa Filho CFF (2020) Breast lesion classification in ultrasound images using deep convolutional neural network. IEEE Access 8:133349–133359
122. Lin C-H, Lai H-Y, Chen P-Y, Wu J-X, Pai C-C, Su C-M et al (2022) Breast lesions screening of mammographic images with 2D spatial and 1D convolutional neural network-based classifier. Appl Sci 12(15):7516
123. Rampun A, López-Linares K, Morrow PJ, Scotney BW, Wang H, Ocaña IG et al (2019) Breast pectoral muscle segmentation in mammograms using a modified holistically-nested edge detection network. Med Image Anal 57:1–17
124. Alkhaleefah M, Tan T-H, Chang C-H, Wang T-C, Ma S-C, Chang L et al (2022) Connected-SegNets: a deep learning model for breast tumor segmentation from X-ray images. Cancers 14(16):4030
125. Chouhan N, Khan A, Shah JZ, Hussnain M, Khan MW (2021) Deep convolutional neural network and emotional learning based breast cancer detection using digital mammography. Comput Biol Med 132:104318
126. Qin C, Lin J, Zeng J, Zhai Y, Tian L, Peng S et al (2022) Joint dense residual and recurrent attention network for DCE-MRI breast tumor segmentation. Comput Intell Neuroscice. 2022:3470764
127. Li L, Tian H, Zhang B, Wang W, Li B (2022) Prediction for distant metastasis of breast cancer using dynamic contrast-enhanced magnetic resonance imaging images under deep learning. Comput Intell Neurosci 2022:126061
128. Xiao X, Gan F, Yu H (2022) Tomographic ultrasound imaging in the diagnosis of breast tumors under the guidance of deep learning algorithms. Comput Intell Neurosci 2022:9227440
129. Akselrod-Ballin A, Chorev M, Shoshan Y, Spiro A, Hazan A, Melamed R et al (2019) Predicting breast cancer by applying deep learning to linked health records and mammograms. Radiology 292(2):331–342
130. Eroğlu Y, Yildirim M, Çinar A (2021) Convolutional Neural Networks based classification of breast ultrasonography images by hybrid method with respect to benign, malignant, and normal using mRMR. Comput Biol Med 133:104407
131. Byra M, Galperin M, Ojeda-Fournier H, Olson L, O'Boyle M, Comstock C et al (2019) Breast mass classification in sonography with transfer learning using a deep convolutional neural network and color conversion. Med Phys 46(2):746–755
132. Samala RK, Chan H-P, Hadjiiski LM, Helvie MA, Cha KH, Richter CD (2017) Multi-task transfer learning deep convolutional neural network: application to computer-aided diagnosis of breast cancer on mammograms. Phys Med Biol 62(23):8894

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133. Li S, Wei J, Chan H-P, Helvie MA, Roubidoux MA, Lu Y et al (2018) Computer-aided assessment of breast density: comparison of supervised deep learning and feature-based statistical learning. Phys Med Biol 63(2):025005
134. Samala RK, Chan H-P, Hadjiiski L, Helvie MA, Richter CD, Cha KH (2018) Breast cancer diagnosis in digital breast tomosynthesis: effects of training sample size on multi-stage transfer learning using deep neural nets. IEEE Trans Med Imaging 38(3):686–696
135. Aboutalib SS, Mohamed AA, Berg WA, Zuley ML, Sumkin JH, Wu S (2018) Deep learning to distinguish recalled but benign mammography images in breast cancer screeningdeep learning in mammography. Clin Cancer Res 24(23):5902–5909
136. Zhang Y, Chen J-H, Chang K-T, Park VY, Kim MJ, Chan S et al (2019) Automatic breast and fibroglandular tissue segmentation in breast MRI using deep learning by a fully-convolutional residual neural network U-Net. Acad Radiol 26(11):1526–1535
137. Wu N, Phang J, Park J, Shen Y, Huang Z, Zorin M et al (2019) Deep neural networks improve radiologists' performance in breast cancer screening. IEEE Trans Med Imaging 39(4):1184–1194
138. Zhou J, Zhang Y, Chang KT, Lee KE, Wang O, Li J et al (2020) Diagnosis of benign and malignant breast lesions on DCE-MRI by using radiomics and deep learning with consideration of peritumor tissue. J Magn Reson Imaging 51(3):798–809
139. Nair AA, Washington KN, Tran TD, Reiter A, Bell MAL (2020) Deep learning to obtain simultaneous image and segmentation outputs from a single input of raw ultrasound channel data. IEEE Trans Ultrason Ferroelectr Freq Control 67(12):2493–2509
140. Mutasa S, Chang P, Van Sant EP, Nemer J, Liu M, Karcich J et al (2020) Potential role of convolutional neural network based algorithm in patient selection for DCIS observation trials using a mammogram dataset. Acad Radiol 27(6):774–779
141. Ha R, Chang P, Karcich J, Mutasa S, Van Sant EP, Liu MZ et al (2019) Convolutional neural network based breast cancer risk stratification using a mammographic dataset. Acad Radiol 26(4):544–549
142. Byra M, Jarosik P, Szubert A, Galperin M, Ojeda-Fournier H, Olson L et al (2020) Breast mass segmentation in ultrasound with selective kernel U-Net convolutional neural network. Biomed Signal Process Control 61:102027
143. Zhang Y, Chen J-H, Lin Y, Chan S, Zhou J, Chow D et al (2021) Prediction of breast cancer molecular subtypes on DCE-MRI using convolutional neural network with transfer learning between two centers. Eur Radiol 31:2559–2567
144. Lu S-Y, Wang S-H, Zhang Y-D (2022) SAFNet: A deep spatial attention network with classifier fusion for breast cancer detection. Comput Biol Med 148:105812
145. Kumar V, Webb JM, Gregory A, Denis M, Meixner DD, Bayat M et al (2018) Automated and real-time segmentation of suspicious breast masses using convolutional neural network. PLoS ONE 13(5):e0195816
146. Zhang C, Zhao J, Niu J, Li D (2020) New convolutional neural network model for screening and diagnosis of mammograms. PLoS ONE 15(8):e0237674
147. Heenaye-Mamode Khan M, Boodoo-Jahangeer N, Dullull W, Nathire S, Gao X, Sinha G et al (2021) Multi-class classification of breast cancer abnormalities using deep convolutional neural network (CNN). PLoS ONE 16(8):e0256500
148. Zhang F-Z, Lin C-H, Chen P-Y, Pai N-S, Su C-M, Pai C-C et al (2022) Number of convolution layers and convolution kernel determination and validation for multilayer convolutional neural network: case study in breast lesion screening of mammographic images. Processes 10(9):1867
149. Zhu M, Pi Y, Jiang Z, Wu Y, Bu H, Bao J et al (2022) Application of deep learning to identify ductal carcinoma in situ and microinvasion of the breast using ultrasound imaging. Quant Imaging Med Surg 12(9):4633–4646
150. Truhn D, Schrading S, Haarburger C, Schneider H, Merhof D, Kuhl C (2019) Radiomic versus convolutional neural networks analysis for classification of contrast-enhancing lesions at multiparametric breast MRI. Radiology 290(2):290–297
151. Yala A, Lehman C, Schuster T, Portnoi T, Barzilay R (2019) A deep learning mammography-based model for improved breast cancer risk prediction. Radiology 292(1):60–66
152. Yala A, Schuster T, Miles R, Barzilay R, Lehman C (2019) A deep learning model to triage screening mammograms: a simulation study. Radiology 293(1):38–46
153. Zhou L-Q, Wu X-L, Huang S-Y, Wu G-G, Ye H-R, Wei Q et al (2020) Lymph node metastasis prediction from primary breast cancer US images using deep learning. Radiology 294(1):19–28
154. Arora R, Rai PK, Raman B (2020) Deep feature–based automatic classification of mammograms. Med Biol Eng Compu 58:1199–1211
155. Sivanandan R, Jayakumari J (2021) A new CNN architecture for efficient classification of ultrasound breast tumor images with activation map clustering based prediction validation. Med Biol Eng Compu 59:957–968
156. El Adoui M, Drisis S, Benjelloun M (2020) Multi-input deep learning architecture for predicting breast tumor response to chemotherapy using quantitative MR images. Int J Comput Assist Radiol Surg 15:1491–1500
157. Fujioka T, Kubota K, Mori M, Kikuchi Y, Katsuta L, Kasahara M et al (2019) Distinction between benign and malignant breast masses at breast ultrasound using deep learning method with convolutional neural network. Jpn J Radiol 37:466–472

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158. Huang H, Chen H, Xu H, Chen Y, Yu Q, Cai Y et al (2021) Cross-tissue/organ transfer learning for the segmentation of ultrasound images using deep residual U-Net. J Med Biol Eng 41:137–145
159. Xiang H, Huang Y-S, Lee C-H, Chien T-YC, Lee C-K, Liu L et al (2021) 3-D Res-CapsNet convolutional neural network on automated breast ultrasound tumor diagnosis. Eur J Radiol. 138:109608
160. Huo L, Hu X, Xiao Q, Gu Y, Chu X, Jiang L (2021) Segmentation of whole breast and fibroglandular tissue using nnU-Net in dynamic contrast enhanced MR images. Magn Reson Imaging 82:31–41
161. Ren T, Lin S, Huang P, Duong TQ (2022) Convolutional neural network of multiparametric MRI accurately detects axillary lymph node metastasis in breast cancer patients with pre neoadjuvant chemotherapy. Clin Breast Cancer 22(2):170–177
162. Witowski J, Heacock L, Reig B, Kang SK, Lewin A, Pysarenko K et al (2022) Improving breast cancer diagnostics with deep learning for mri. Sci Transl Med 14(664):eabo4802
163. Gao F, Wu T, Li J, Zheng B, Ruan L, Shang D et al (2018) SD-CNN: A shallow-deep CNN for improved breast cancer diagnosis. Comput Med Imaging Graph 70:53–62
164. Daoud MI, Abdel-Rahman S, Bdair TM, Al-Najar MS, Al-Hawari FH, Alazrai R (2020) Breast tumor classification in ultrasound images using combined deep and handcrafted features. Sensors 20(23):6838
165. Rehman KU, Li J, Pei Y, Yasin A, Ali S, Mahmood T (2021) Computer vision-based microcalcification detection in digital mammograms using fully connected depthwise separable convolutional neural network. Sensors 21(14):4854
166. Tsai K-J, Chou M-C, Li H-M, Liu S-T, Hsu J-H, Yeh W-C et al (2022) A high-performance deep neural network model for BI-RADS classification of screening mammography. Sensors 22(3):1160
167. Shia W-C, Hsu F-R, Dai S-T, Guo S-L, Chen D-R (2022) Semantic segmentation of the malignant breast imaging reporting and data system lexicon on breast ultrasound images by using DeepLab v3+. Sensors 22(14):5352
168. Yin H, Bai L, Jia H, Lin G (2022) Noninvasive assessment of breast cancer molecular subtypes on multiparametric MRI using convolutional neural network with transfer learning. Thoracic Cancer 13(22):3183–3191
169. Zhou K, Li W, Zhao D (2022) Deep learning-based breast region extraction of mammographic images combining pre-processing methods and semantic segmentation supported by Deeplab v3+. Technol Health Care 13:1–18
170. Wang Y, Wang N, Xu M, Yu J, Qin C, Luo X et al (2019) Deeply-supervised networks with threshold loss for cancer detection in automated breast ultrasound. IEEE Trans Med Imaging 39(4):866–876
171. Shu X, Zhang L, Wang Z, Lv Q, Yi Z (2020) Deep neural networks with region-based pooling structures for mammographic image classification. IEEE Trans Med Imaging 39(6):2246–2255
172. AlGhamdi M, Abdel-Mottaleb M, Collado-Mesa F (2020) DU-Net: Convolutional network for the detection of arterial calcifications in mammograms. IEEE Trans Med Imaging 39(10):3240–3249
173. Rahimpour M, Saint Martin M-J, Frouin F, Akl P, Orlhac F, Koole M et al (2022) Visual ensemble selection of deep convolutional neural networks for 3D segmentation of breast tumors on dynamic contrast enhanced MRI. Eur Radiol 3:1–11
174. Zhou J, Luo LY, Dou Q, Chen H, Chen C, Li GJ et al (2019) Weakly supervised 3D deep learning for breast cancer classification and localization of the lesions in MR images. J Magn Reson Imaging 50(4):1144–1151
175. Qiu Y, Yan S, Gundreddy RR, Wang Y, Cheng S, Liu H et al (2017) A new approach to develop computer-aided diagnosis scheme of breast mass classification using deep learning technology. J Xray Sci Technol 25(5):751–763
176. Gu J, Wang Z, Kuen J, Ma L, Shahroudy A, Shuai B et al (2018) Recent advances in convolutional neural networks. Pattern Recogn 77:354–377
177. Li Z, Liu F, Yang W, Peng S, Zhou J (2021) A survey of convolutional neural networks: analysis, applications, and prospects. In: IEEE transactions on neural networks and learning systems
178. Yoon Y, Swales G Jr, Margavio TM (1993) A comparison of discriminant analysis versus artificial neural networks. J Oper Res Soc 44(1):51–60
179. Lawrence J (1993) Introduction to neural networks. California Scientific Software
180. Bengio Y, Goodfellow I, Courville A (2017) Deep learning. MIT press Cambridge, MA, USA
181. Li H, Gong M, Zhang M, Wu Y (2021) Spatially self-paced convolutional networks for change detection in heterogeneous images. IEEE J Selected Topics Appl Earth Observ Remote Sens 14:4966–4979
182. Howarth R (2017) Analysis, matrix transpose, orthogonal matrix, singular-value decomposition, skew-symmetric matrix, symmetric matrix
183. Wolf K (2013) Integral transforms in science and engineering. Springer Science & Business Media, New York
184. Quintard M, Whitaker S (1994) Transport in ordered and disordered porous media I: the cellular average and the use of weighting functions. Transp Porous Media 14:163–177
185. Jie HJ, Wanda P (2020) RunPool: a dynamic pooling layer for convolution neural network. Int J Comput Intell Syst 13(1):66–76
186. Sainath TN, Vinyals O, Senior A, Sak H, Eds. (2015) Convolutional, long short-term memory, fully connected deep neural networks. In: 2015 IEEE international conference on acoustics, speech and signal processing (ICASSP). IEEE
187. Tao Y, Gao X, Hsu K, Sorooshian S, Ihler A (2016) A deep neural network modeling framework to reduce bias in satellite precipitation products. J Hydrometeorol 17(3):931–945

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188. Yi H, Shiyu S, Xiusheng D, Zhigang C, Eds. (2016) A study on deep neural networks framework. In: 2016 IEEE Advanced Information Management, Communicates, Electronic and Automation Control Conference (IMCEC). IEEE
189. Alzubaidi L, Duan Y, Al-Dujaili A, Ibraheem IK, Alkenani AH, Santamaría J et al (2021) Deepening into the suitability of using pre-trained models of ImageNet against a lightweight convolutional neural network in medical imaging: an experimental study. PeerJ Computer Science 7:e715
190. Doegar A, Dutta M, Gaurav K (2019) Cnn based image forgery detection using pre-trained alexnet model. Int J Comput Intell & IoT 2(1):6
191. Wen L, Li X, Li X, Gao L, Eds. (2019) A new transfer learning based on VGG-19 network for fault diagnosis. In: 2019 IEEE 23rd international conference on computer supported cooperative work in design (CSCWD). IEEE
192. Budhiman A, Suyanto S, Arifianto A, Eds (2019) Melanoma cancer classification using resnet with data augmentation. In: 2019 International Seminar on Research of Information Technology and Intelligent Systems (ISRITI). IEEE
193. Zhang M, Li W, Du Q (2018) Diverse region-based CNN for hyperspectral image classification. IEEE Trans Image Process 27(6):2623–2634
194. Thuan D (2021) Evolution of Yolo algorithm and Yolov5: The State-of-the-Art object detention algorithm
195. Badrinarayanan V, Handa A, Cipolla R (2015) Segnet: A deep convolutional encoder-decoder architecture for robust semantic pixel-wise labelling. arXiv preprint arXiv:150507293
196. Sarmad M, Lee HJ, Kim YM, Eds (2019) Rl-gan-net: A reinforcement learning agent controlled gan network for real-time point cloud shape completion. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition
197. He Y, Yao S, Yang W, Yan H, Zhang L, Wen Z et al (2021) An extraction method for glacial lakes based on Landsat-8 imagery using an improved U-Net network. IEEE J Select Topics Appl Earth Observ Remote Sens 14:6544–6558
198. Hooley RJ, Scoutt LM, Philpotts LE (2013) Breast ultrasonography: state of the art. Radiology 268(3):642–659
199. Kang M, Kwon H, Park J-H, Kang S, Lee Y (2020) Deep-asymmetry: asymmetry matrix image for deep learning method in pre-screening depression. Sensors 20(22):6526
200. Ekert T, Krois J, Meinhold L, Elhennawy K, Emara R, Golla T et al (2019) Deep learning for the radiographic detection of apical lesions. J Endod 45(7):917–225.e5
201. Lee CT, Kabir T, Nelson J, Sheng S, Meng HW, Van Dyke TE et al (2022) Use of the deep learning approach to measure alveolar bone level. J Clin Periodontol 49(3):260–269
202. Benjelloun M, El Adoui M, Larhmam MA, Mahmoudi SA, Eds. (2018) Automated breast tumor segmentation in DCE-MRI using deep learning. In: 2018 4th International Conference on Cloud Computing Technologies and Applications (Cloudtech). IEEE
203. Tandel GS, Biswas M, Kakde OG, Tiwari A, Suri HS, Turk M, Laird JR, Asare CK, Ankrah AA, Khanna NN, Khanna BK (2019) A review on a deep learning perspective in brain cancer classification. Cancers 11(1):111
204. Cuesta ABC, Ríos MDM, Meseguer MRN, Velasco JAG, de Matías MM, Sotillos SB et al (2019) Accuracy of tumor size measurements performed by magnetic resonance, ultrasound and mammography, and their correlation with pathological size in primary breast cancer. Cirugía Española (English edition) 97(7):391–396
205. Pereira S, Pinto A, Alves V, Silva CA (2016) Brain tumor segmentation using convolutional neural networks in MRI images. IEEE Trans Med Imaging 35(5):1240–1251
206. Goodfellow I, Bengio Y, Courville A (2016) editors. Deep learning MIT press (2016). In: Conference on information and communication systems (ICICS)
207. Qin C, Wu Y, Zeng J, Tian L, Zhai Y, Li F et al (2022) Joint transformer and multi-scale CNN for DCE-MRI breast cancer segmentation. Soft Comput 26(17):8317–8334
208. Wadhwa A, Bhardwaj A, Verma VS (2019) A review on brain tumor segmentation of MRI images. Magn Reson Imaging 61:247–259
209. Bouraghi H, Rezayi S, Amirazodi S, Nabovati E, Saeedi S (2022) Evaluating the usability of a national health information system with heuristic method. J Edu Health Promot 11(1)
210. Barton D, Woetzel J, Seong J, Tian Q (2017) Artificial intelligence: implications for China
211. Shahmoradi L, Safdari R, Mirhosseini MM, Rezayi S, Javaherzadeh M (2023) Development and evaluation of a clinical decision support system for early diagnosis of acute appendicitis. Sci Rep 13(1):19703
212. Carriero A, Groenhoff L, Vologina E, Basile P, Albera M (2024) Deep learning in breast cancer imaging: state of the art and recent advancements in early 2024. Diagnostics 14(8):848
213. Seah JC, Tang CH, Buchlak QD, Holt XG, Wardman JB, Aimoldin A et al (2021) Effect of a comprehensive deep-learning model on the accuracy of chest x-ray interpretation by radiologists: a retrospective, multireader multicase study. The Lancet Digital Health 3(8):e496–e506

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